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The Changing Nature of Firm Innovation: Short-Termism and Influential Innovation in U.S. Public Firms

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Abstract. We examine the link between short-term pressures and technologically significant innovation in U.S. public firms in 1997–2015. Using a market-based measure of short-term pressure, we estimate its relationship with influential and novel patents. We find that firms facing more intense short-term pressures are less likely to patent highly influential or novel innovations. To evaluate whether this relationship is causal, we use changes in ownership styles following financial institution mergers as instruments. Our analysis suggests that changing short-term pressures from investors had a causal impact on firm innovative outcomes; this finding is robust to a wide variety of empirical specifications. While public firms as a whole retained a constant share of highly influential patents, this activity has become more concentrated in fewer firms. This shift does not appear to be fully compensated by an increase in technologically significant patents by nonpublic firms such as venture-capital (VC)-backed start-ups. These findings raise questions about capital markets' impact on firm R&D strategy and the nature of innovative activities in public firms.

History: Accepted by Ashish Arora, entrepreneurship and innovation.

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Keywords: short-termism • R&D • patent influence • innovation • institutional investors

1. Introduction

Long-term corporate research strategies yielded transformative innovations, including DuPont's development of nylon and AT&T's Bell Labs invention of the transistor. DuPont's decades-long research program on nylon transformed the company into a fibers-dominant entity by the 1950s (Ndiaye 2007), while Bell Labs' transistor research, spanning over a decade, proved so groundbreaking that eight of its researchers received Nobel Prizes.[1] These foundational innovations disrupted technological trajectories in chemicals and electronics, contributing significantly to the U.S. economy for decades and generating higher valuations for firms engaged in such innovations (Arora et al. 2024a). Despite these achievements, both DuPont and AT&T have subsequently abandoned the R&D strategies that enabled these discoveries in favor of short-term, incremental approaches focused on leveraging existing knowledge. This shift raises a broader question about whether similar pressures have affected innovation strategies across firms in the economy. In this paper, we examine whether increasing short-term capital market pressures on U.S. public firms shifted their innovation strategies away from generating technologically significant and novel patents.

The prolonged development period required for major innovations poses a fundamental challenge to long-term R&D investment. Development is inherently time-intensive because successful projects must resolve interconnected technical problems that emerge sequentially, requiring years or decades to complete (Rosenberg 1969, Arthur 2009, Goldfarb and Kirsch 2025, Kim et al. 2025). Despite the long-term growth and industry impact these innovations have historically fueled, investors often lack

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patience for delayed returns and pressure firms to shift away from more foundational R&D. For example, DuPont came under assault in 2015 from activist shareholder Nelson Peltz, who sought a much more short-term-oriented strategy, proposing to shut down DuPont's central research labs in favor of more immediate returns. Although Peltz failed, the 2016 Dow-DuPont merger resulted in the shutdown of its R&D division.[2]

How pervasive are stories like DuPont's? More generally, has a documented increase in short-term pressures on firms from capital markets (Sampson and Shi 2023) led firms to retreat from technologically significant innovation? Although firm R&D expenditures increased from $30.5B in 1980 to $259.9B in 2013 (National Science Foundation 2024), there has been a significant shift away from research toward development of more applied technical knowledge.[3] Concerns about the aggregate impact of such a shift have raised concerns that a new division of labor, which relies more heavily on universities, technology markets, and small firms, has not adequately compensated for the reduction in fundamental corporate research (Arora et al. 2020; see also Bernstein 2015, Budish et al. 2015, Knott 2016 for additional documentation of shifting R&D priorities among firms).[4]

We use several approaches to explore our central research question: Did increased short-term pressures on firms make it more likely that they developed less technologically significant patent portfolios (conditional on patenting)? We begin by documenting several patterns based on U.S. patent-based measures that capture technological significance from 1985 through 2009 or 2015, depending on data availability. We measure technological significance in two distinct ways using conceptually distinct measures, namely: (1) the forward-looking network of direct and indirect forward citations (i.e., patent influence (Corredoira and Banerjee 2015)) and (2) novelty of firm patents based on original word combinations relative to prior art (i.e., lexicographic or text-based patent novelty (Arts et al. 2021)), within a technology class in a given year.[5] With these measures, we capture technologically significant patents produced by public firms, universities and federal labs, non-venture-capital (VC)-backed private firms, and VC-backed private firms.

Our documented patterns reveal three central facts. First, public and private firms' share of the most influential and novel patents has remained relatively constant throughout the period. Although the university and private-VC-backed share has increased considerably relative to their own share in 1985, this increase remains small relative to the share of public and other private firms. Second, the seemingly stable group proportions conceal trends within groups. Since around the year 2000, generation of high-impact patents among public firms has undergone significant concentration, with technologically significant innovations increasingly originating from a progressively smaller subset of companies. Third, within public firms, the production of technologically significant patents is negatively related to short-term pressures on firms.

Prior work shows an increase in average short-term orientation among U.S. public firms (Sampson and Shi 2023). Using the market-based measure of short-termism in U.S. public firms documented by these authors, we establish that, at the firm level, more short-term-oriented firms are less likely to generate technologically significant patents. We do not find evidence that this reduction in technologically significant patents is made up for by acquisitions of innovative small private firms.

To address the endogeneity of market-based short-termism measures, we employ an instrumental variable (IV) approach to establish a causal link between changing short-term pressures and technologically significant innovation. Our instruments leverage shifts in institutional investor trading behavior at the portfolio level that alter portfolio-wide short-term pressures independent of firm-specific characteristics. Specifically, we examine changes in institutional investor holding periods following investor mergers as our instrumental variables, because these post-merger portfolio-wide adjustments are likely uncorrelated with individual firms' technological prospects or unique attributes. Changes in institutional investor trading patterns influence stock prices and, consequently, managerial behavior through stock-based compensation mechanisms, even when driven by factors unrelated to firm fundamentals.

Our analyses consistently suggest that, for the average firm, increased short-term pressures contribute to reduced production of influential and novel inventions, a finding that is robust across multiple specifications and empirical approaches (as detailed in our extensive Online Appendix and referenced below). This finding takes on particular significance in the current innovation landscape, where R&D productivity has stagnated despite increased expenditure (Knott 2016), the net benefit of patenting has declined (Kogan et al. 2017), and public firms' publication output has not kept pace with patent production (Arora et al. 2020).

These findings have significant implications for corporate strategy and policy. Whereas prior research has documented a shift in corporate R&D investment patterns toward more incremental projects, our study demonstrates that this shift has stemmed directly (at least in part) from increased financial market pressures on firms to prioritize short-term returns. By establishing this causal link, we show how capital market dynamics have effectively reshaped firms' innovation strategies, driving firms toward lower-risk, shorter-term initiatives despite the potential for significant long-term value creation through fundamental research. This market-driven reorientation of corporate innovation has broad implications for public market participation, innovation policy design, and

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industrial organization, particularly as firms weigh the trade-offs between public capital market access and maintaining long-term research autonomy.

The paper proceeds as follows. We begin with existing research documenting rising short-term orientation among firms and the implications for firm investment, followed by a conceptual discussion of what drives technologically significant inventions in Section 2. Our data, key measures, and empirical approach are described in Section 3. Section 4 sets out results, starting with trends in influential and novel patents, which are our proxies for technologically significant inventions, over time. We follow with our focal empirical results that demonstrate a persistent negative correlation between short-term pressures and technologically significant innovation as well as an instrumental variable approach that offers causal evidence for the relationship between short-termism and innovation. Section 5 concludes.

2. Patient Capital, Firm Investment, and the Nature of Innovation

We first establish that short-term orientation is increasing among public firms and why we might expect this to impact investment behavior in Section 2.1. In Section 2.2, we explore how this may tie to the production of technologically significant innovations.

2.1. Short-Term Orientation and Firm Investment Behavior

Rising short-term orientation, that is, a focus on short-term over longer-term payoffs from investments, and the impact of that change on investment behaviors have been documented in several prior studies. Davies et al. (2014) and Sampson and Shi (2023) relate investor short-term pressures to firm strategies. Davies et al. (2014) highlight how managers, influenced by investors who excessively discount future returns, prioritized immediate dividends over long-term investments. This result challenges traditional financial theories and reveals a trend of growing investor preference for near-term returns in the United States and United Kingdom. Sampson and Shi (2023) document a similar trend in U.S. public markets, showing a shift toward shorter firm time horizons and its correlation with various firm-level strategic factors that have been argued to reflect short-term firm behavior. Both studies underscore the significant implications this shift has for firm investment behaviors and broader economic growth.

Other studies provide a more nuanced understanding of this phenomenon. For example, Bushee (1998) documents that firms exhibited more myopic investment behavior by reducing R&D spending when their shares were held by "transient" institutional investors known for high portfolio turnover and momentum trading.[6] This shift in investment priorities was likely a strategic firm response to avoid earnings decreases and subsequent share price declines.[7] Complementing these findings, Flammer and Bansal (2017) and Desjardine (2015) observe that firms with a long-term orientation, often captured via long-term CEO compensation packages, were more inclined toward investments with extended payoff periods, such as R&D or other durable investments.[8] Asker et al. (2015) further illustrate this by documenting that firms transitioning from private to public ownership typically reduced their overall investments and became less responsive to investment opportunities, driven by the inherent pressures for short-term returns in public markets.

CEOs may influence strategic decisions beyond just changing R&D spending levels. For example, Cummings and Knott (2018), Islam and Zein (2020), and Harrison et al. (2024) find that CEO differences influenced strategic direction within investment categories, even after accounting for investment levels. For example, Sanjay Mehrotra stated that, as CEO of Sandisk, he evaluated technological capabilities and challenges, engaged in strategic planning, and prioritized critical issues in R&D (Islam and Zein 2020, p. 506). This influence over R&D direction may be particularly pronounced when a CEO is a founder of the company and is directly involved in the development of key technologies, as seen in the case of Bill Gates at Microsoft (Harrison et al. 2024). Furthermore, the Cummings and Knott (2018) case analysis of C. Michael Armstrong, the first outside CEO of Hughes Aircraft Company, illustrates how Armstrong influenced the company's R&D outcomes by changing project selection criteria and restructuring R&D units. Together, these studies emphasize that pressures on CEOs, whether from external market forces or internal organizational incentives, can shape not only their discretion over funding decisions but also the specific focus areas of R&D, determining both innovation direction and overall research productivity.[9]

In addition to those highlighted above, there are two specific mechanisms through which short-term orientation likely constrains disruptive technology development. First, firm researchers face significant challenges in communicating the value of novel projects to managers and investors with compressed evaluation horizons, because these projects require extended periods for fuller assessment of their potential. Second, even when breakthrough ideas are well understood, their development timelines frequently extend beyond the planning horizons that short-term-oriented management teams typically allow for. The following section provides illustrations of these mechanisms in more detail.

2.2. Conditions for Technologically Significant Innovation

To link short-term market pressure, management priorities, and the nature of innovation, we examine why

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the innovative process of technologically significant inventions—that is, those that change technological trajectories in a meaningful way—may differ from that of technologically incremental innovations. Innovation that changes technological trajectories typically involves the creation of multiple additional technologies to solve the original problem that the invention seeks to address. When these inventions are novel in that they build less on established technologies, organizations will be required to venture into unexplored technical territory to make them functional. Further, the research process of improving novel inventions typically reveals a host of additional technological problems that need to be solved—interdependent challenges that resist quick solutions.[10] Given this inherent complexity and uncertainty in developing trajectory-shifting innovations, conditions conducive to incremental innovations and more technologically significant innovations will differ, with the latter emphasizing patience while initial solutions are built out into fully functional systemic solutions.

Building on Rosenberg (1963, 1969), Arthur (2009) creates a structured framework for the innovation process, leveraging the observation that technological innovation is recombinatory (Fleming 2001) and hierarchical. Inventors focus on an existing concept or idea, and work to solve a series of technological challenges needed to render the idea operational. Early efforts to convert an idea into a working technology, that is, an artifact that *does* something, usually lead to prototypes that are not particularly useful. However, such efforts are still fruitful because they bring to light problems that still need to be solved; that is, they identify what the barriers to implementation are. In this way, efforts to solve each problem reveal a set of subproblems, the solutions to which, if and when they are worked out, are embodied in technologies and subtechnologies, respectively. When ideas are novel in the sense that they rely on a new principle to accomplish a task, such as using a transistor as opposed to a vacuum tube to amplify, we expect that it will take more time to work out how subtechnologies fit together to create an overall solution (see, e.g., Kim et al. (2025) and Goldfarb and Kirsch (2025) for examples of this innovative process in the context of bionic prosthetics and aviation, respectively).

Arthur (2009) illustrates these ideas via the example of Gary Starkweather's invention of the laser printer at Xerox in late 1969. Starkweather came up with the fundamental idea while working in Xerox's laboratory in Webster, NY. The Webster laboratory was organized with a sharp focus on incremental improvements that could be embedded in the next version of Xerox copiers. Starkweather's laser idea, which came out of his doctoral work, recombined the (then newly invented) laser to "'paint' an image on a xerographic drum" (Arthur 2009, p. 117). When Starkweather proposed using laser technology to produce a printer in the laboratory, it was quickly recognized that creating a working laser printer was not possible by simply assembling off-the-shelf components (Hiltzik 2000). Reflecting the strategic priorities of the Webster laboratory, Starkweather was ordered to refocus his efforts on marginal improvements to existing, diffuse-light, cathode-ray tube (CRT) machines. Unhappy with this directive, and with a bit of luck, Starkweather was transferred to the newly created Xerox Palo Alto Research Center (PARC). PARC was organized completely differently, akin to an academic department with a long-term focus. With the freedom at PARC, Starkweather had both time and resources to solve an array of subproblems of laser printing. This led to the 1969 breakthrough patent that was technologically significant and influenced many later innovations. Thus, the laser printer was invented in an environment largely free from short-term pressure, with fewer restrictions on budgets and freedom to work on interesting projects. This illustration suggests how a firm's temporal preferences and patience for returns can impact the *nature* of technological investments and outcomes.[11]

In contrast to the laser printer example, Arthur (2009) considers inventions incremental when necessary subtechnologies are already well established, as in the case of marginal improvements to CRT copiers. To develop these more incremental innovations, short-term horizons likely suffice because it is possible to map specific improvements directly to product performance and customer value. That is, an incremental innovation might be thought of as a solution to a smaller subset of problems, whereas when an idea is more novel, it generally requires solutions to a host of new, unanticipated subtechnology problems. Thus, novel ideas are not simply more time-consuming, they are riskier because they require additional solutions to subtechnological problems. These new problems are difficult to foresee when the original idea is imagined, and each could be very difficult to solve. As such, for firms and their investors to generate and realize value from significant inventions that solve foundational problems not yet considered or imagined by anyone, they likely need great patience for returns amid uncertain conditions and considerable time commitments.

These insights have direct implications for our choice of measures of technological significance, described in detail below. At times, the key important advance will be the application of a new principle to a problem. Such innovations are likely to use new words to describe their solutions, and appear as lexicographically novel. However, though such an innovation may resolve an important bottleneck or signal a new approach, there are often additional bottlenecks that need to be resolved to make the innovation useful. For

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example, the first-generation germanium solid-state transistor demonstrated a principle but had limited functionality. The substitution of germanium with silicon was necessary for the first commercial application and subsequent patents built on this silicon-based invention, even though the initial germanium-based transistor was more novel (Riordan 2007). Innovations that solve bottlenecks and create useful technological trajectories typically generate dense citation trees as developers build functional systems around these foundational advances (Goldfarb and Kirsch 2025). This pattern of patent features leads us to consider different measures of technological significance based on text-based novelty, citation-based influence, and their combined effect. Thus, we infer that firms producing relatively few technologically significant inventions are likely conducting limited exploration of fundamental principles, instead focusing on more incremental innovation spaces.

In this sense, firms facing pressures to deliver quick returns may find it difficult to commit to more novel and/or influential R&D projects because these projects typically require solutions to a larger set of subtechnological problems, some of which only become salient and known sequentially, and thus are characterized by greater uncertainty and longer time horizons (Goldfarb and Kirsch 2025). Instead, these firms may prefer simpler projects—that is, those that involve more straightforward applications of current technological solutions, because these projects will more likely lead to near-term, more-certain payoffs. Indeed, others have speculated that short-termism may have contributed to the declining support for R&D projects with long horizons (Arora et al. 2020), such as those focused on scientific advancements (Arora et al. 2018). Although previous studies have examined the relationship between short-termism and R&D investments, the findings have been inconclusive (e.g., Roe 2021, 2022). We still lack comprehensive evidence and understanding regarding how short-termism shapes the fundamental nature and technological significance of innovative outputs, beyond any effects on R&D investment levels.

These arguments lead us to the central prediction of the paper: *Firms facing more short-term pressure from investors are less likely to produce technologically significant inventions.*

3. Research Design

To demonstrate and explore the relationship between short-term orientation and technologically significant innovation, we begin by documenting several central empirical patterns. First, relying on prior work, we note the validity of our proxy for short-term pressure and map its increase over time. We then document patterns in the technological significance of innovative output across different organizational types, including U.S. public and private firms, the concentration of technologically significant patents among U.S. public firms, and the relationship between significant innovation and firm short-term orientation. To evaluate whether this relationship is causal, we then employ an instrumental variable approach, leveraging variation in investor ownership behavior patterns that is plausibly exogenous to firms' technological opportunities to evaluate subsequent changes in innovative outcomes.

To evaluate the robustness of our findings, we engage in a two-pronged approach. First, we explore the robustness of the documented correlations to a broad range of specifications using epistemic maps (King et al. 2021). These maps, detailed in Online Appendix B, demonstrate that the central correlation (between short-term orientation and technologically significant innovation) is robust across a wide range of empirical specifications. Second, we explore the robustness of our IV estimation to violations of the exclusion restriction of our instrumental variables. This leverages recently developed "partial identification" techniques which quantify how severe the violation must be to affect our inference (Conley et al. 2012, Frake et al. 2025).

Ultimately, our empirical objective is to evaluate whether exogenous shifts in short-term orientation impact firm innovation. Thus, we consider two central explanations for the relationship between rising short-term orientation and declining influential inventions: (1) that our observed correlations are driven by market trends exogenous to the firm (i.e., changes in investor preference that are unrelated to firm characteristics or opportunities) and/or (2) that the correlation is a result of evolving opportunity sets for firms, influenced by the changing nature of technology (including the possibility that advancements in the technological landscape have made long-term innovation more challenging than before, or that the types of organizations engaging in such long-term innovation have changed). Our identification strategy identifies the former explanation and assumes the latter explanation is also operative.

3.1. Sample Description

To assess whether firms with short time horizons are associated with less influential inventions, we bring together firm-level patent and financial data with a market-based measure of short-term orientation. Our primary estimation sample includes firms publicly traded on major U.S. stock exchanges (i.e., NYSE/AMEX and NASDAQ) during the period 1997–2009. This sampling frame accommodates the time windows and the diverse data sources required for constructing the measures (see Online Appendix E for details) while ensuring the availability of all covariates used in the analysis. For analyses of the text-based patent novelty measure, which does not

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require a seven-year forward time window as the influence measure does, we extend the sample period through 2015. Note that the construction of our short-term orientation measure requires at least five consecutive years of stock market data for each firm. Consequently, we exclude firms that were publicly listed for fewer than five consecutive years.[12] As noted below, we construct a panel data set with one observation per firm for each year that firm appears in the sample subject to these data constraints.

3.2. Dependent Variables Capturing Technologically Significant Inventions

To characterize the nature of firm innovation, we measure the overall influence of a firm's inventions on subsequent innovations (Corredoira and Banerjee 2015) and the text-based novelty of firm inventions relative to prior innovations (Arts et al. 2021).[13] We use patent grant and application information collected from PATSTAT Global, the National Bureau of Economic Research (NBER) data set, and PatentsView. Our sample includes all U.S. Patent and Trademark Office (USPTO) applications for utility patents with application dates between January 1, 1976, and December 31, 2016, and granted before July 31, 2018. With these data, we calculate influence for patents filed during the period 1980–2009 and text-based novelty for patents filed during the period 1980–2015. These variables are calculated according to the following definitions and are aggregated annually at the firm portfolio level.[14]

Patent Influence captures a focal patent's direct and indirect impact over future technological developments (Corredoira and Banerjee 2015, Corredoira et al. 2018). We first calculate the patent-level influence as

$$Patent\ Influence = \sum_{k=1}^{\infty} (\alpha^k A^{Tk})\, e \tag{1}$$

where α is the attenuation factor (in this study, $\alpha = 1$), k is the citation generation, A^T is the transpose of the adjacency matrix defined by patent citations, and e is the vector representing whether the patent application is ultimately granted ($e = 1$) or not ($e = 0$).[15] Intuitively, this measure captures how many generations of patent trees are built upon a focal patent. In other words, the influence of a patent is the column sum of the matrix $\alpha^k A^{Tk}$. The citation tree defining the adjacency matrix is composed of all direct and indirect citations received by the focal patent from patents granted before 2018 and with a filing date in the seven-year window following the publication date of the focal patent. Patents, unlike academic literature, cite only their immediate antecedents, creating a de facto barrier to assessing influence beyond the direct antecedents. Therefore, capturing indirect citations with patent data is particularly important to identify expansion into new frontiers. *Patent Influence* reflects how much the focal patent has contributed to subsequent inventions, with larger scores reflecting greater long-term utilization of the knowledge embedded in a focal patent. Because the distribution of the original values of *Patent Influence* has very long right tails, we follow Corredoira and Banerjee (2015) and Corredoira et al. (2018) and take the natural log of the measure plus one.[16]

Over our sample time frame, there have been systemic changes in patenting regimes and an overhaul of the patenting system that, among other things, resulted in different rules for the attribution of priority (i.e., the first filing date for a patent application) and the publication of application documents. The implementation of these changes resulted in an increase in the number of citations to patent applications, which impacts our citation-based measures of granted patents. Citations to applications were negligible before 2000, but by the 2010s, well above 50% of patent citations were to applications rather than granted patents. In addition to this change, patents also show systematic differences in the average number of citations over time and across classes.

For these reasons, we normalize our patent variables of interest to account for field and time differences. We standardize our measures using a z-score approach, constructed by dividing the focal patent's influence by the standard deviation of all patents' influence in the focal patent's primary class in a given year.[17] We do not subtract the average or median from the z-scores, because this approach allows a more natural interpretation given that citation measures all have a zero lower bound. Therefore, our measure reflects the number of standard deviations from zero in a given year. With this approach, our measure is comparable across patent classes and years. The influence variables were derived from the dataset published by Corredoira and Goldfarb (2025).

Formally, for each patent class and year, we calculate the standard deviation of patent influence, and then the z-score of each firm patent:

$$Patent\ Influence\ (z\ score)_i = \frac{Patent\ Influence_i}{Std.\ Dev.\ (Patent\ Influence_j)} \tag{2}$$

where $Std.\ Dev.\ Patent\ Influence_j$ is the standard deviation of the patent influence of all patents filed in a given year within the focal patent i's class j (i.e., Cooperative Patent Classification (CPC)). As a final step, we aggregate the measure to the firm level by calculating the mean of the z-scores for all patents in a firm's portfolio in a filing year and log the measure plus one due to skewness.

Patent Novelty is a text-based measure that identifies novel word combinations utilized by a patent, because such words are thought to identify patents that are the result of more exploratory R&D strategies (Balsmeier

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et al. 2018). We use a patent novelty measure introduced by Arts et al. (2021), who identify keyword combinations that appear for the first time in patent titles, abstracts, and claims. The novelty measure is backward-looking and allows us to extend the sample period to 2015 for estimations. We first take a logged count of the number of novel pairwise keyword combinations plus one for each patent. Similar to the influence measure above, we then calculate the *z*-score for this novelty measure by patent class and grant year to adjust for differences across fields and time periods. Finally, to generate a firm-level measure, we take the average *z*-scores for the firm's patent portfolio by filing year.[18] Thus, our measures of *Patent Influence* and *Patent Novelty* capture how much a firm's patent portfolio varies from the patent influence or novelty of other patents in the same classes and year. This measurement approach naturally controls for widespread technology shifts that impact many firms and ensures that our dependent variables are capturing truly technologically significant innovation, whether measured via influence or novelty.

We test our central prediction using both citation-based and text-based outcomes, as these measures leverage distinct types of data, address each other's limitations, and offer complementary perspectives. Based on our discussion above, *Patent Influence* is derived from the forward citation network of the focal patent and thus is forward-looking. It reflects the realized significance of a patent by capturing its direct and indirect impact on future technologies. In this sense, a patent with high influence has solved a problem that is useful in a large number of additional innovations, and thus technologically important. In contrast, *Patent Novelty* evaluates the originality of the text description of a patent compared with prior patents, to capture more novel approaches to solving problems. The measure offers a retrospective assessment of the novel insights in the focal patent, capturing its linguistic contributions that represent departures from prior technologies.[19] When novel approaches solve major bottlenecks, they too can be thought of as technologically important. The production of novel or influential patents may, therefore, correspond to distinct corporate strategies and choices. Consequently, we examine how short-term orientation relates to each type of innovation output. In certain cases, an innovation will both be very novel and also generate substantial follow-on activities if subsequent bottlenecks to applications are solved quickly (i.e., high influence). To capture this possible intersection, *Influence* × *Novelty* is defined as the product of a firm's technological influence and novelty.

We assign the measures by the year of patent application because this represents the earliest observation of the invention. Capturing application year is particularly important for a citation-based measure because patent applications are now published in advance of grant (for patents applied for from 2000 onward), making it possible for a patent to receive citations before it is granted. Although we only include patents that are later granted in our data set, we utilize all citations received within a seven-year window of publication of the application or, where the application is not observed, the grant.[20]

3.3. Focal Independent Variable: Short-Term Orientation (Implied Discount Rate (*IDR*))

As a proxy for short-term pressures on firms, we use a measure employed by Sampson and Shi (2023) that estimates an implied discount rate used in the market's valuation of firm assets and expected future cash flows. The measure is derived from an asset pricing model first proposed by Miles (1993) and later used by Davies et al. (2014). Essentially, we estimate the rate at which investors discount the expected future cash flows of a firm above and beyond the firm-specific risk premium. The rationale is that a shorter time horizon (i.e., stronger pressure for short-term returns) leads investors and firms to focus heavily on near-term returns and place less emphasis on more distant returns; this effectively results in a much steeper discount rate for cash flows in later periods. A detailed explanation of how and why this measure captures a firm's short-term orientation, as well as a more thorough evaluation of robustness and alternative explanations, is the subject of Sampson and Shi (2023) and not reproduced here. However, we note salient aspects of this measure here.

Conceptually, we use a dividend discount model and treat stock price of firm j at time t as the net present value of its future cash flows over the next N periods.[21] The model, which is presented as equation (1) in Sampson and Shi (2023), can thus be written as

$$P_{jt} = \frac{E_t[D_{jt+1}]x_{jt}}{(1 + r_t + \pi_{jt})} + \frac{E_t[D_{jt+2}]x_{jt}^2}{(1 + r_t + \pi_{jt})^2} + \cdots \frac{(E_t[D_{jt+N}] + E_t[P_{jt+N}])x_{jt}^N}{(1 + r_t + \pi_{jt})^N} + \varepsilon \quad (3)$$

We adopt Sampson and Shi's (2023, p. 235) variable definitions: "P_{jt} is the stock price, $E_t[D_{jt+N}]$ is the expected value of firm j's dividends in future period $t + N$, r_t is the market risk-free rate, π_{jt} is the firm's risk premium, and N is the number of time periods used in the estimation of the current stock price." $E_t[P_{jt+N}]$ is the expected value of the terminal stock price at the end of the time window, $t + N$. Following equation (2) in Sampson and Shi (2023), we define the firm-specific risk premium as

$$\pi_{jt} = \alpha_1 \beta_{jt} + \alpha_2 Z_{jt}. \quad (4)$$

Using this specification (Sampson and Shi 2023, p. 238), β_{jt} is firm j's beta, and Z_{jt} is the firm's gearing (i.e., debt/equity) in period t. These factors capture a

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firm's risks associated with price volatility and leverage. α_1, α_2 are estimated coefficients associated with these factors.

The firm-specific, time-varying parameter x is thus an additional discount rate that is not explained by the firm-specific risk premium and captures how much the market values future cash flows of the firm. If there is no significant discounting (or overvaluing) above and beyond the sum of the risk-free rate and the firm-specific risk premium, then x should equal one in theory; this is the underlying null hypothesis of the model. Empirically, most estimates of x are close to one, offering validating support for the model. However, we also observe that they vary meaningfully between and within firms over time.

We follow the empirical estimation procedures detailed in Sampson and Shi (2023, pp. 237–238): " … we use current dividends and the current stock price to proxy for the current expectation of future dividends and stock price … We obtain the company beta from CRSP, which calculates annual betas for public companies using the methods set out in Scholes and Williams (1977). A firm's debt and equity are obtained from COMPUSTAT."[22] Substituting these proxies and Equation (5) into Equation (4), we obtain the following equation for the empirical estimation of a nonlinear random coefficient model (equation (3) in Sampson and Shi 2023):

$$P_0 = \frac{(D_0)x_0}{(1 + r_0 + \alpha_1\beta_0 + \alpha_2 Z_0)} + \frac{(D_0)x_0^2}{(1 + r_0 + \alpha_1\beta_0 + \alpha_2 Z_0)^2} + \cdots \frac{(D_{j0} + P_{j0})x_0^N}{(1 + r_0 + \alpha_1\beta_0 + \alpha_2 Z_0)^N} + \varepsilon \quad (5)$$

Sampson and Shi (2023) estimate several alternative versions of Equation (5) above, some of which change the calculation of the firm-specific risk premium to include additional risk factors identified by Fama and French (1992).[23] Because results on these alternative specifications substantively match those used in this paper, but significantly reduce the sample size, we focus on the results from estimating Equation (5) above. A firm's short-term orientation, our independent variable of interest, is an implied discount rate (*IDR*), $1 - x$. We subtract x from one to ease interpretation of results; that is, a larger value of *IDR* indicates increased short-term pressures on a firm as implied in its asset price. (Note that we estimate *IDR* from five years of a firm's securities data, as set out in our data set construction details in Online Appendix E.)

This approach suggests that the stock price, set by market investors, reflects the implied discounting rate based on a firm's risk profile, expected cash flows, and investor preference for short-term returns. Consequently, at least part of this discount rate is influenced by external market forces rather than solely reflecting internal firm decisions; it is these external forces that we argue impact the firm's R&D strategy. Thus, an increase in *IDR* may lead to a reduced focus on technologically significant innovations, as firms adjust their R&D horizons in response to shifting market-imposed time pressures. In our instrumental variable analysis below, we isolate the elements of *IDR* that are exogenous to firm characteristics to identify a more causal relationship between short-term pressures facing a firm and the technological significance of its later patents. In Online Appendix C, we provide a comprehensive set of supplementary analyses designed to explore and demonstrate the robustness of the *IDR* measure to variations in underlying model assumptions, as well as to evaluate the sensitivity of its empirical link with firm innovation. These additional tests support the validity and robustness of the empirical link between *IDR* and firm innovation.

3.4. Control Variables

To account for relevant firm characteristics that may explain any observed relationship between *IDR* and our measures of technologically significant innovation, we include several control variables that we expect might be correlated with *IDR* and our dependent variables. Although we set out variable definitions and data sources in Table 1, we briefly discuss the rationale for inclusion of these variables here.[24]

Greater *Financial Slack* may be associated with a longer horizon and also a greater ability to invest in long-term R&D projects. *R&D Intensity* may be positively associated with *IDR* and would clearly be expected to be associated with more innovation.[25] *Firm Patent Portfolio Size* is a general measure of innovative activity and strategic patenting; as such, it should be associated with future innovative activity absent marked changes in firm strategy.[26] *Market Competition* measures the degree to which a firm's goods and services are in a competitive space, which is plausibly associated with both *IDR* and innovative outcomes. For instance, firms may explore radical innovation to escape the competitive pressure of close rivals (Aghion et al. 2005). This is a continuous, time-varying measure of the similarity of goods and services between firms based on text analysis of firms' 10K filings (Hoberg and Phillips 2010).

To capture the R&D differences between single-business and multibusiness companies, we include a dummy for *Multibusiness Firm*. Given the presence of agency and coordination issues that may hinder innovation alongside learning benefits that may enhance it, there is no clear consensus on whether multibusiness firms are more innovative or not (Nelson 1959, Porter 1992, Seru 2014). However, multibusiness firms may be

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Table 1. Control and Instrumental Variable Descriptions

Variable	Data source	Description
Financial Slack	Compustat	Five-year moving average of the difference between current assets and liabilities, divided by total firm assets
R&D Intensity	Compustat	Five-year moving average of R&D spending scaled by total firm assets
Firm Patent Portfolio Size	Kogan et al. (2017)	Logged term of the number of patents a firm filed in a year based on firm identifier "permno" in the KPSS data
Market Competition	Hoberg and Phillips data library	Sum of the similarity score for all competitors of the firm in the same product market, as defined by the Text-based Network Industry Classifications and derived from text analysis of firm 10K product descriptions (Hoberg and Phillips 2010). The market competition variable is only measured from 1995 onwards. Including or excluding the pre-1995 period does not meaningfully impact our main regression results
Multibusiness Firm	Compustat	Dummy that equals one if the firm has more than one business segment
Firm Size	Compustat	Logged term of the total assets of the firm
Transition from Dedicated after Merger	Bushee (1998); He and Huang (2017)	The number of existing institutional owners of the focal firm that shift from dedicated ownership in the same year or the year following a merger, within the past two years
Transition from Transient after Merger	Bushee (1998); He and Huang (2017)	The number of existing institutional owners of the focal firm that shift from transient ownership in the same year or the year following a merger, within the past two years

less centralized with less centralized R&D, which is often more development oriented. In contrast, prior research suggests that more centralized firms' R&D programs are likely to engage in basic research (Arora et al. 2014) and generate outcomes with broader technological impact (Argyres and Silverman 2004).[27] We also include *Firm Size*, despite its ambiguous effect on the nature of innovation. Larger firms may have more significant R&D programs over time and may be able to assume the increased risk surrounding longer-term, more uncertain projects. However, smaller firms may exhibit greater agility and adaptability, making them better suited for highly impactful R&D. Additionally, large firms' focus on improvement of existing core technologies may limit their ability to undertake radical innovations.

Although past research sheds light on appropriate constructions of the above control variables, the theoretical links among short-term orientation, the controls, and innovative output do not clarify the timing in which these effects may operate. Hence, determining the appropriate lag structure for these controls is not obvious. In our tables, we report results with control variables lagged by one year. However, our central results are insensitive to these concerns and lag structures, as we report in the epistemic maps set out in Online Appendix B. Descriptive statistics for all variables are set out in Tables 2 and 3. Table 2 reports descriptive statistics for different samples used in our analyses. Table 3 reports pairwise correlations.

3.5. Empirical Approach and Model Estimation

We employ two primary approaches to estimate the relationship between *IDR* and technologically significant firm innovation: (1) an examination of within-firm changes via fixed-effect ordinary least squares (OLS) regressions (Table 4) and (2) an instrumental variable approach to evaluate whether observed relationships are potentially causal (Table 5). We also set out an epistemic mapping approach in the Online Appendix (Online Appendix B) to evaluate the sensitivity of results from approaches (1) and (2) to varying empirical assumptions that lack clear theoretical guidance (e.g., lag structures and combinations of controls).

4. Empirical Results

This section presents our empirical analyses of the relationship between short-term orientation and technologically significant innovation. We begin by documenting key trends in short-term orientation (*IDR*) and innovation measures (*Patent Influence*, *Patent Novelty*, and *Influence* × *Novelty*) over time, highlighting important patterns across different types of organizations (Section 4.1). We then establish the core relationship between a firm's short-term orientation and the technological significance of its patent portfolio, demonstrating robust negative associations across multiple specifications (Section 4.2). To address potential endogeneity concerns, we employ instrumental variable techniques using institutional investor classification changes following mergers, providing evidence that this negative relationship is causal rather than merely correlational (Section 4.3). Finally, we investigate whether short-term pressures shift the locus of technologically significant innovation and examine how short-term orientation specifically impacts breakthrough innovations that represent substantial technological

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Table 2. Descriptive Statistics by Sample

Variable		Patenting firms (n = 25,661)	Non-patenting firms (n = 22,902)	Influence and novelty sample (n = 7,796)	Novelty sample (n = 14,924)
Patent influence	Mean			1.010	
	Median			0.981	
	Std Dev.			0.535	
	Min			0.000	
	Max			4.059	
Patent novelty	Mean			1.797	1.637
	Median			1.706	1.559
	Std Dev.			3.541	2.708
	Min			0.000	0.000
	Max			300.383	300.383
Influence × Novelty	Mean			1.687	
	Median			1.447	
	Std Dev.			1.450	
	Min			0.000	
	Max			29.562	
IDR	Mean	−0.023	−0.020	−0.032	−0.023
	Median	−0.026	−0.025	−0.032	−0.026
	Std Dev.	0.022	0.025	0.017	0.021
	Min	−0.062	−0.058	−0.062	−0.062
	Max	0.390	0.471	0.372	0.372
Financial Slack	Mean	0.319	0.209	0.335	0.328
	Median	0.306	0.184	0.322	0.314
	Std Dev.	0.232	0.245	0.225	0.234
	Min	−3.770	−12.308	−3.770	−3.770
	Max	0.967	1.000	0.957	0.957
R&D Intensity	Mean	0.092	0.022	0.100	0.102
	Median	0.035	0.000	0.059	0.053
	Std Dev.	0.162	0.095	0.136	0.161
	Min	0.000	0.000	0.000	0.000
	Max	5.126	5.481	2.079	5.481
Firm Patent Portfolio Size	Mean	2.058	0.000	2.395	2.255
	Median	1.609	0.000	2.079	1.792
	Std Dev.	1.885	0.000	1.574	1.538
	Min	0.000	0.000	0.693	0.693
	Max	9.893	0.000	9.091	9.091
Market Competition	Mean	2.960	2.054	3.409	3.439
	Median	0.531	0.588	0.738	0.694
	Std Dev.	5.737	3.641	5.901	6.280
	Min	0.000	0.000	0.000	0.000
	Max	58.360	48.437	33.677	58.360
Multibusiness Firm	Mean	0.485	0.453	0.495	0.496
	Median	0.000	0.000	0.000	0.000
	Std Dev.	0.500	0.498	0.500	0.500
	Min	0.000	0.000	0.000	0.000
	Max	1.000	1.000	1.000	1.000
Firm Size	Mean	6.193	5.560	6.406	6.538
	Median	6.128	5.542	6.348	6.473
	Std Dev.	2.204	2.014	2.130	2.184
	Min	0.533	0.001	0.678	0.340
	Max	12.764	11.784	12.527	12.764

Notes. The novelty sample covers public firms whose patents were filed in 1997–2015 and granted by 2018. The influence and novelty sample includes public firms with patents filed in 1997–2009 and granted by 2011. The narrowing of sampling period is due to the seven-year forward window required by the influence measure and the time lags between patent filing and grants.

advances (Section 4.4). Throughout our analysis, we find consistent evidence that increased short-term orientation is associated with diminished technologically significant innovation, with important implications for firm strategy and policy.

4.1. Trends over Time: *IDR*, *Patent Influence*, and *Patent Novelty*

Sampson and Shi (2023) comprehensively document the short-term orientation measure (*IDR*) at the firm, industry, and market levels. This research demonstrates an

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Table 3. Pairwise Correlations

Influence and novelty sample (n = 7,796)

	Patent Influence	Patent Novelty	Influence × Novelty	IDR	Financial Slack	R&D Intensity	Firm Patent Portfolio Size	Market Competition	Multibusiness Firm	Firm Size
Patent Influence	1.000									
Patent Novelty	0.004	1.000								
Influence × Novelty	0.662	0.183	1.000							
IDR	−0.115	−0.033	−0.113	1.000						
Financial Slack	0.041	0.014	0.064	−0.339	1.000					
R&D Intensity	−0.036	0.032	0.072	−0.205	0.346	1.000				
Firm Patent Portfolio Size	−0.157	−0.027	−0.146	0.088	−0.223	−0.086	1.000			
Market Competition	−0.059	0.046	0.082	−0.237	0.420	0.599	−0.030	1.000		
Multibusiness Firm	−0.066	−0.031	−0.083	0.204	−0.337	−0.286	0.193	−0.281	1.000	
Firm Size	−0.147	−0.021	−0.132	0.255	−0.540	−0.420	0.644	−0.182	0.338	1.000

Patent novelty sample (n = 14,924)

	Patent Novelty	IDR	Financial Slack	R&D Intensity	Firm Patent Portfolio Size	Market Competition	Multibusiness Firm	Firm Size
Patent Novelty	1.000							
IDR	−0.070	1.000						
Financial Slack	0.017	−0.271	1.000					
R&D Intensity	0.034	−0.176	0.281	1.000				
Firm Patent Portfolio Size	−0.031	0.120	−0.182	−0.078	1.000			
Market Competition	0.057	−0.173	0.372	0.558	−0.056	1.000		
Multibusiness Firm	−0.029	0.168	−0.307	−0.280	0.171	−0.291	1.000	
Firm Size	−0.034	0.349	−0.525	−0.405	0.588	−0.194	0.322	1.000

Table 4. IDR and Technological Significance

Dependent variable	Influence		Novelty		Novelty		Influence × Novelty	
Period	1997–2009		1997–2009		1997–2015		1997–2009	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
IDR	−3.9006	−3.1875	−4.8480	−3.6386	−7.8708	−5.8146	−11.3406	−8.9477
	(1.0880)	(0.9589)	(1.4873)	(1.2943)	(0.8259)	(0.8441)	(2.8259)	(2.3485)
Financial Slack		0.1167		0.3478		0.1592		0.4442
		(0.1002)		(0.1392)		(0.0946)		(0.2615)
R&D Intensity		0.0621		0.0148		−0.0987		0.2907
		(0.1393)		(0.1846)		(0.1300)		(0.4011)
Firm Patent Portfolio Size		−0.0382		0.0121		0.0231		−0.0782
		(0.0121)		(0.0154)		(0.0168)		(0.0284)
Market Competition		0.0075		0.0066		0.0036		0.0239
		(0.0033)		(0.0056)		(0.0037)		(0.0093)
Multibusiness Firm		−0.0440		−0.0465		−0.0342		−0.1328
		(0.0202)		(0.0321)		(0.0316)		(0.0547)
Firm Size		−0.0818		−0.1252		−0.1404		−0.2606
		(0.0173)		(0.0239)		(0.0201)		(0.0432)
Intercept	0.8858	1.4749	1.4956	2.1897	1.3640	2.2401	1.3262	3.0658
	(0.0346)	(0.1351)	(0.0473)	(0.1823)	(0.0193)	(0.1530)	(0.0898)	(0.3445)
N (observations)	7,796	7,796	7,796	7,796	14,924	14,924	7,796	7,796
R^2 (within)	0.013	0.027	0.009	0.019	0.010	0.013	0.0160	0.0346

Notes. OLS models with firm fixed effects. The unit of analysis is firm-year. Robust standard errors clustered on firms are reported in parentheses. *Influence* captures the logged average size of the forward citation trees of focal patents filed by a focal firm in a given year. *Novelty* captures the average amount of new word combinations in the patents filed by the firm in a year. These outcomes are constructed as z-scores, which are normalized within the patent class and grant year and thus control for time and field differences. *Influence* × *Novelty* is the product of *Influence* and *Novelty*. Explanatory variables are defined in Table 1 and lag the dependent variable by one year. Models (1)–(4) and Models (7) and (8) are based on the influence and novelty sample, which covers firms with patents filed in years 1997–2009 and granted through 2011. Models (5) and (6) are based on the novelty sample, which includes firms with patents filed in 1997–2015 and granted by 2018.

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Table 5. IV Analysis with Mergers—IDR and Technological Significance

	Patent Influence		Patent Novelty		Influence × Novelty	
Dependent variable	Stage 1 IDR	Stage 2 Influence	Stage 1 IDR	Stage 2 Novelty	Stage 1 IDR	Stage 2 Influence × Novelty
IDR		−7.0125		−5.3227		−13.3619
		(1.9461)		(1.9417)		(5.2391)
Financial Slack	−0.0075	0.0333	−0.0019	0.1645	−0.0075	0.3000
	(0.0065)	(0.1216)	(0.0039)	(0.1160)	(0.0065)	(0.2889)
R&D Intensity	0.0093	0.2596	0.0334	−0.1445	0.0093	0.9025
	(0.0144)	(0.1790)	(0.0079)	(0.2006)	(0.0144)	(0.4022)
Firm Patent Portfolio Size	0.0014	−0.0522	0.0020	−0.0588	0.0014	−0.2114
	(0.0006)	(0.0159)	(0.0004)	(0.0139)	(0.0006)	(0.0397)
Market Competition	−0.0005	0.0031	−0.0005	−0.0026	−0.0005	0.0138
	(0.0002)	(0.0037)	(0.0002)	(0.0045)	(0.0002)	(0.0099)
Multibusiness Firm	0.0011	−0.0240	0.0011	−0.0359	0.0011	−0.0712
	(0.0009)	(0.0227)	(0.0009)	(0.0380)	(0.0009)	(0.0570)
Firm Size	0.0056	−0.0575	0.0127	−0.1432	0.0056	−0.2031
	(0.0009)	(0.0240)	(0.0007)	(0.0298)	(0.0009)	(0.0607)
Transition from Dedicated after Merger	−0.0111		−0.0142		−0.0111	
	(0.0005)		(0.0008)		(0.0005)	
Transition from Transient after Merger	−0.0031		−0.0069		−0.0031	
	(0.0006)		(0.0004)		(0.0006)	
N (observations)		5,665		10,764		5,665
F-test of excluded instruments	*F*(2, 996)	226.20	*F*(2, 1,414)	230.88	*F*(2, 996)	226.20
Prob > *F*		0.0000		0.0000		0.0000

Notes. 2SLS models with firm fixed effects. The unit of analysis is firm-year. Robust standard errors clustered on firms are reported in parentheses. *Influence* captures the logged average size of the forward citation trees of focal patents filed by a focal firm in a given year. *Novelty* captures the average amount of new word combinations in the patents filed by the firm in a year. These outcomes are constructed as *z*-scores, which are normalized within the patent class and grant year and thus control for time and field differences. *Influence* × *Novelty* is the product of *Influence* and *Novelty*. *Transition from Dedicated (Transient) after Merger* is defined as the number of existing institutional owners of the focal firm that shift away from dedicated (transient) ownership in the same year or the year following an institution merger, measured over the two years before current year $t + 1$, and the instrumented *IDR* is measured over year $t + 1$ through $t + 5$ in stage 1. For stage 2, all covariates lag the dependent variable by one year as the main analysis. We focus on acquiring institutions in the mergers because of data availability and drop firms whose institutional owners have never gone through a merger during the sampling period from the analysis.

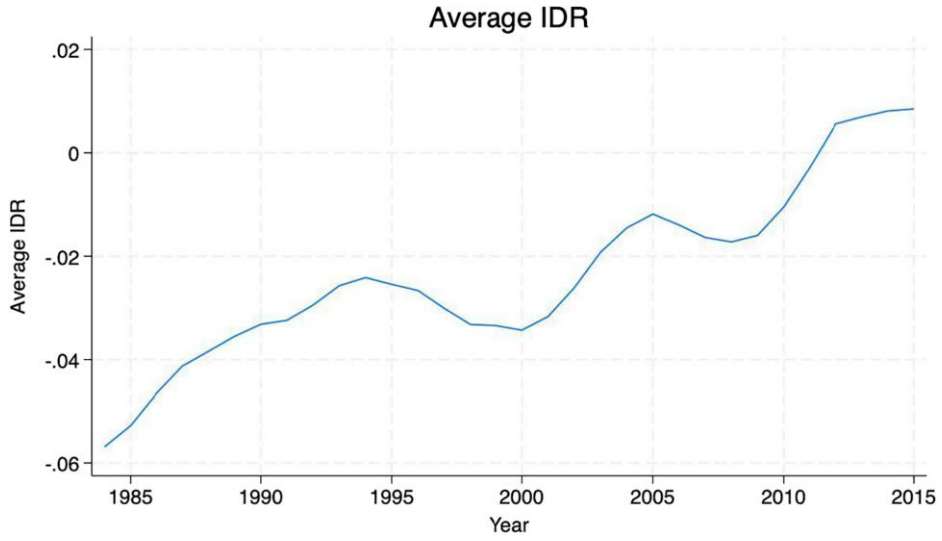
upward trajectory in average *IDR* across firms over time, with notable variation between industries and among firms within the same sector. Our analysis of sample firms yields consistent findings, as illustrated in Figure 1. Both our visualization and Sampson and Shi's (2023) study indicate that *IDR* has increased for most public firms during the observed period.

To illustrate trends in technologically significant innovation, we graph three key trends over time: (1) the distribution of the most technologically significant patents across different assignee categories (e.g., public firm, private firm, etc.), (2) the concentration of these patents among U.S. public firms, and (3) the relationship between a firm's short-term orientation and its holdings of technologically significant patents.

Figure 2, (a) and (b), graphs the proportion of technologically significant patents held by assignee type, including public and private firms, from 1985–2009. We capture patents with either *Patent Influence* or *Patent Novelty* at or above the 95th percentile of *Patent Influence* or *Patent Novelty* (i.e., top 5%) for that year. The analysis of patent distribution patterns reveals several important trends regarding technological influence and novelty across different organizational types. When examining the proportion of top 5% patents by technological influence and novelty, U.S. organizations—including public firms, private firms, universities, and venture-capital-backed start-ups—have maintained relatively stable shares over time. However, foreign firms have shown an increasing share of novel patents during the study period. Although there is no major reallocation in the distribution of technologically significant patents among most types of U.S. organizations, such as between public and private firms, some notable exceptions exist. Universities have seen their share of novel patents grow from 1.8% to 4.2%, whereas VC-backed firms have increased their holdings of influential patents from 0.5% to 5.3% between 1985 and 2009, indicating subtle yet meaningful changes in the innovation ecosystem over time.[28] Overall, these patterns are consistent with some changes in division of labor in innovative activity (Rosen 1991), as well as the documented shortcomings of universities and small firms in overcoming the decline in scientific research in large firms (Arora et al. 2024a).

Further investigation into the concentration of technologically significant innovation reveals a striking trend. Figure 3, (a) and (b), shows a marked increase in

Figure 1. (Color online) Mapping Average IDR, 1985–2015



Note. The graph shows the yearly average *IDR* across sample firms (i.e., U.S. publicly listed firms), from 1985–2015, with increasing *IDR* values indicating greater short-term orientation.

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the proportion of influential and novel patents held by the top four firms, rising from 22% in 1985 to over 44% in 2009. Note that these top four firms are necessarily included in all other groupings represented, because these measurements reflect concentrations. This trend suggests a narrowing in the locus of technologically significant innovation, with such advances becoming increasingly concentrated among a smaller number of firms. This pattern aligns with previous arguments by prior researchers, including Gordon (2017), Bloom et al. (2020), and Knott (2016), who argue that innovation is becoming more challenging and cost-intensive, thus constraining it to a smaller number of well-resourced organizations.

Figure 4, (a) and (b), shows the relationship between firms' short-term orientation and their production of technologically significant innovation. Firms experiencing the highest short-term pressures consistently hold the smallest number and proportion of the most influential or novel patents. Conversely, firms facing the least short-term pressure maintain the largest number of the most influential patents. Organizations experiencing low to moderate short-term pressures demonstrate similar levels of novel patent holdings. Although these observations do not establish causality or account for other potential factors influencing technological significance in patent portfolios, they provide prima facie support for our detailed estimation results set out below.

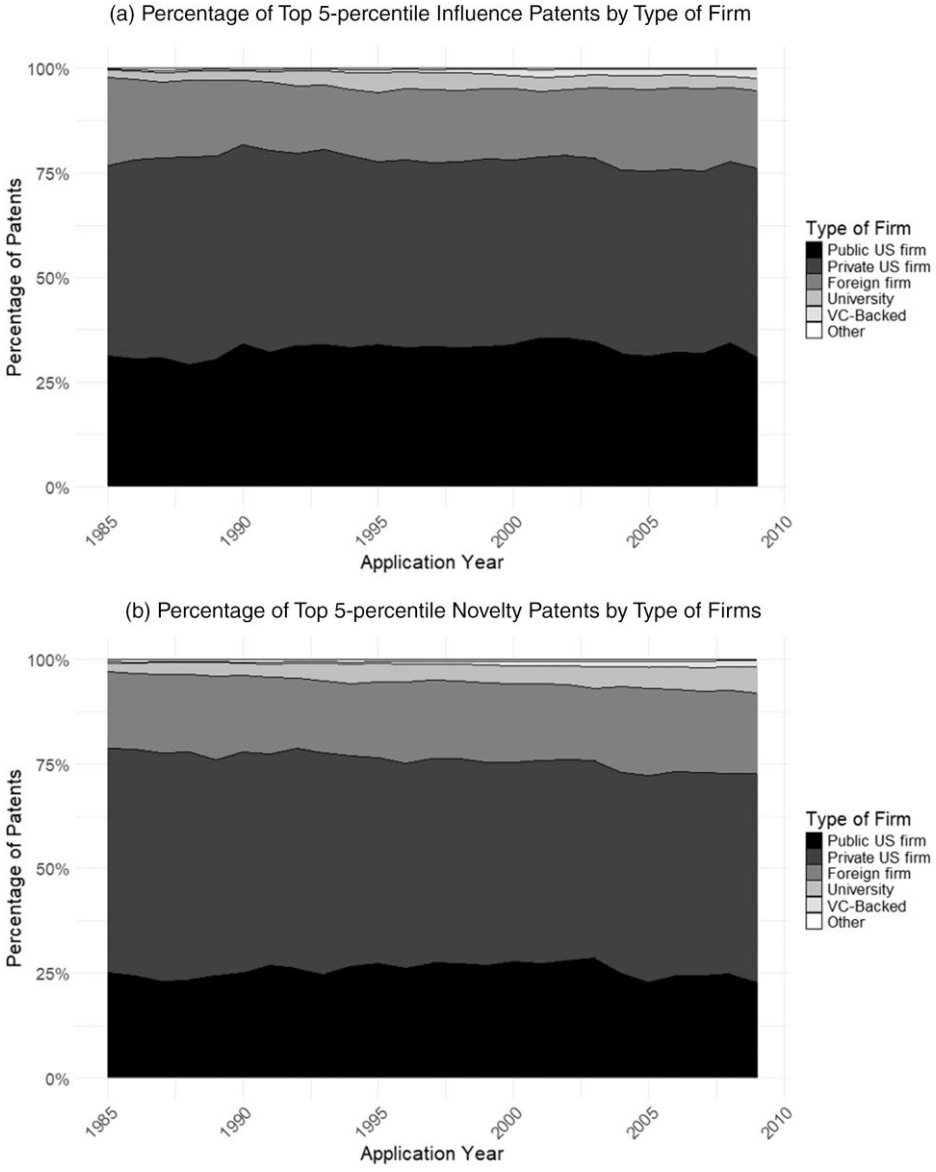
Figure 5, (a) and (b), plots the raw data across time periods, to better visualize the relationship between short-term orientation (*IDR*) and technologically significant patent portfolios over time. Each point shown in these figures represents a firm's average influence or novelty for its patent portfolio and short-term orientation (*IDR*) in a specific time period: 1997–2002, 2003–2009 (influence only), or 2003–2015 (novelty only). Both panels show that short-term pressures have increased over time (i.e., the "cloud" of observations shifts to the right in the latter period), as well as declining technological significance over time (i.e., the cloud shifts downward in the latter period). This negative relationship, which is plotted without adjusting for any controls, is also shown by the fitted lines for each period. The slopes for the post-2002 periods appear more negative than the earlier period. This is consistent with the descriptives shown as well as our analyses below, suggesting that the relationship between short-termism and technologically significant innovation has strengthened over time.

4.2. Relationship Between Short-Term Orientation (*IDR*) and the Influence and Novelty of Firm Innovation

Table 4 presents our central results estimating the relationship between *IDR* and our measures of technologically significant firm patent portfolios. Columns (1) through (4) use identical sampling using a firm's average portfolio *Patent Influence* and *Patent Novelty* as key dependent variables. Because our sample measuring *Patent Novelty* runs until 2015, we also reestimate with this larger sample to evaluate potential changes over time. Finally, columns (7) and (8) display results with *Influence* × *Novelty* (i.e., the product measure between average *Patent Influence* and average *Patent Novelty*).[29] These linear models all include firm fixed effects and,

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Figure 2. Proportion of Highly Influential and Novel Patents by Assignee Group



Notes. The figure shows the percentage of patents in the top fifth percentile of influence (a) and novelty (b), held within the assignee type group (noted in the legend), based on application year. Influence is based on the tree of direct and indirect forward citations, whereas novelty is based on a text-based metric relative to prior art. Each stacked area plot displays the relative contribution of six organizational sectors: public U.S. firms, private U.S. firms, foreign firms, universities, VC-backed entities, and others. Note that foreign firms include foreign entities that do not otherwise appear in the other listed categories. Because we do not have complete data on the firms in the rest of the world, these firms may be publicly listed on non-U.S. exchanges or private firms domiciled outside the United States. The graph demonstrates that, although there are some modest changes over time in the proportions held by each of the assignee groups, there are not significant changes in the assignee type holding the most influential patents. Although foreign firms appear to have increased their share of the most novel patents held over time, other changes among assignee groups are more modest and do not show a reordering of groups holding the most novel patents.

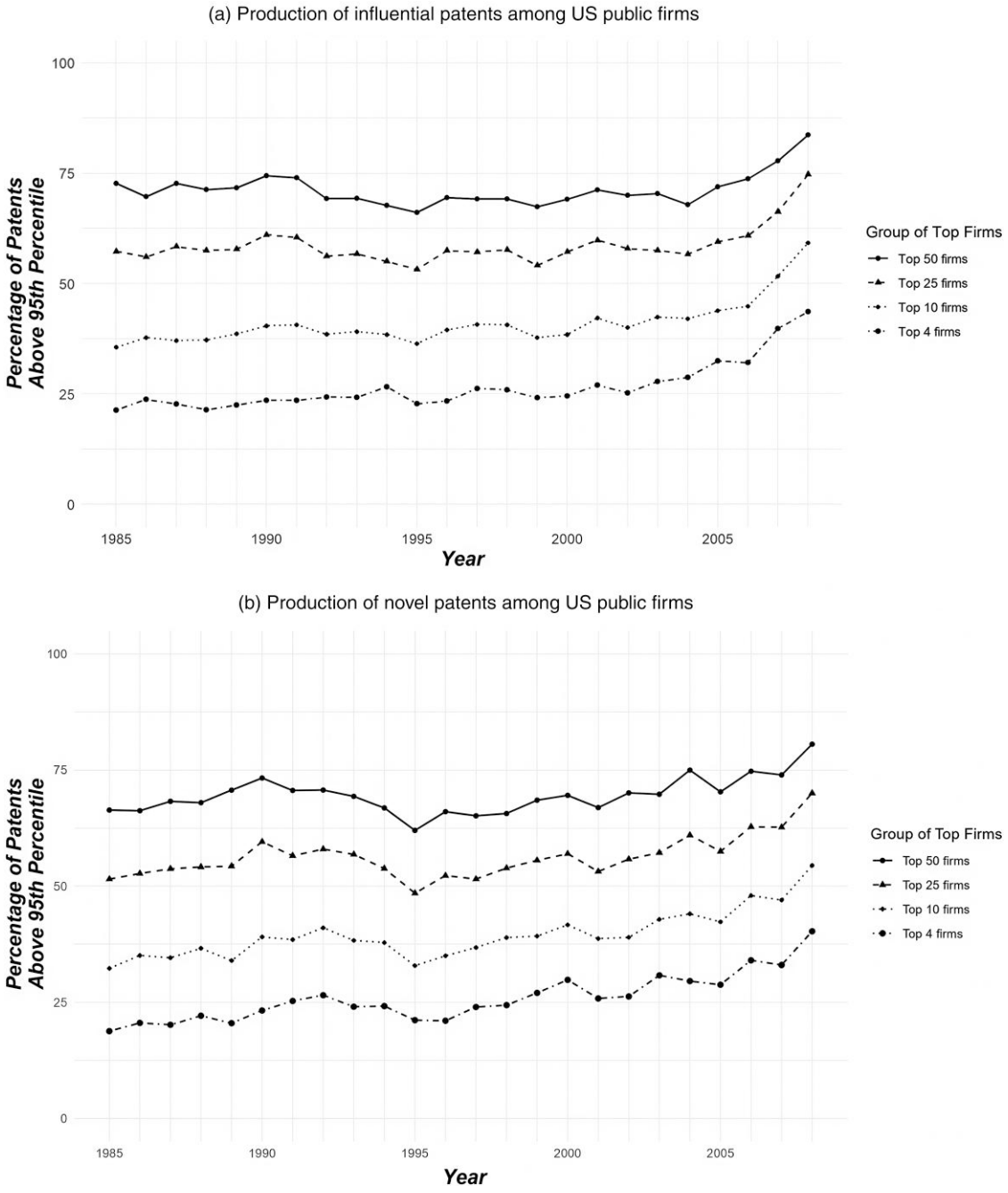
because all dependent variables are normalized by patent class and year (i.e., *z*-scores), control for time and field differences.

Results are consistent across model specifications and document a robust negative relationship between *IDR* and all dependent variables capturing technologically significant innovation. The coefficient on *IDR* is negative and precisely estimated for all models (*p*-values are 0.002 or smaller).[30] Although not reported in the paper, we note that the findings remain highly consistent when we use the traditional measure of direct forward citations to firm patents (i.e., patent impact), thus indicating that rising *IDR* is associated with declines in both direct and indirect citations.

To illustrate the economic magnitude of the results, we examine how the technological significance of a firm's patent portfolio shifts with a one-standard-deviation increase in *IDR*. Given the *IDR* standard

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Figure 3. Concentration of Influential and Novel Patents Among U.S. Public Firms



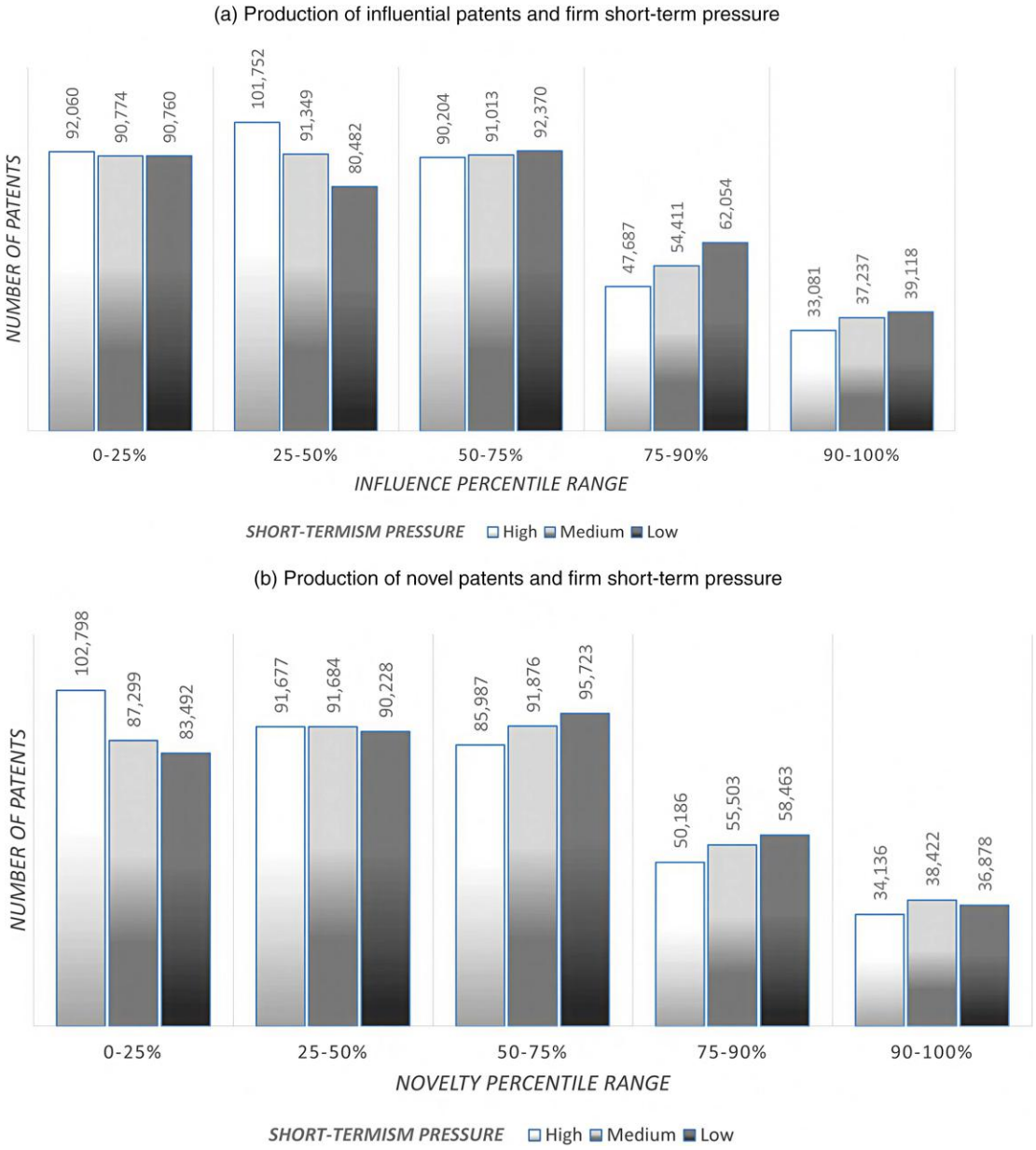
Notes. The figure shows the percentage of the top fifth percentile of influential (a) and novel (b) patents held by U.S. public firms. Influence is based on the tree of direct and indirect forward citations, whereas novelty is based on a text-based metric relative to prior art. Top 4 (10/25/50) firms refers to the four (10/25/50) U.S. public firms in that year that have the highest percentage of influential or novel patents. From 1985 until 2009, the number of U.S. public firms engaged in research that produces highly influential or novel patents is declining, as evidenced by the increase in the percentage of top 5% influential or novel patents held by, for example, the top four firms.

deviation of approximately 0.02 in the patent influence sample, a one-standard-deviation increase in *IDR* corresponds to a 0.07 decrease in *Patent Influence*, which represents a 6.9% decline relative to the sample mean of 1.01. Intuitively, this indicates that the patents in a sample firm's portfolio are 6.9% less influential when compared with an average patent in the same class and year. These results reflect general patterns in the data; firms with *IDR* values in the bottom quartile have patent portfolios with significantly higher *Patent Influence* and *Patent Novelty* than those in the top quartile on average ($p < 0.001$).

Thus, firms that face more short-term pressures for near-term returns, captured via *IDR*, have patent portfolios that are, on average, less influential, less novel,

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Figure 4. (Color online) Short-Termism (*IDR*), Patent Influence, and Patent Novelty in U.S. Public Firms

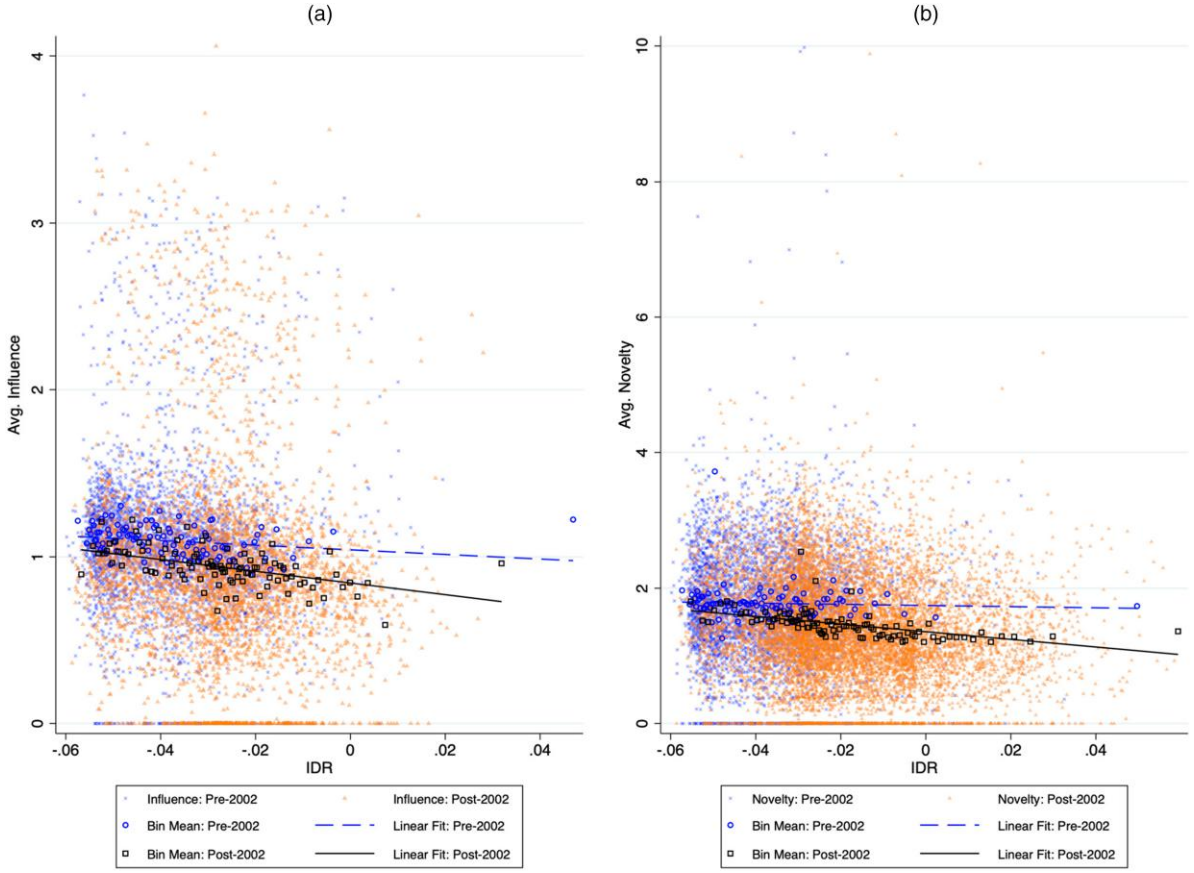


Notes. The figure shows the number of patents in each percentile range of influence (a) or novelty (b) held by U.S. public firms in each tercile of short-termism (low/medium/high *IDR*). Influence is based on the tree of direct and indirect forward citations, whereas *novelty* is based on a text-based metric relative to prior art. There are significantly fewer of the most novel and influential patents at all levels of short-termism. Further, firms facing the highest short-term pressures hold the lowest number and proportion of the most influential and novel patents.

and less technologically significant. A within-sample illustration of this relationship is A. Schulman Inc., a specialty plastics manufacturer targeted by activist hedge fund Barington Capital Group in 2005. Barington cited prolonged stock underperformance and threatened a forced sale. In response to this investor pressure, Schulman enacted cost-cutting and margin improvement measures. Compared with the period before the activist takeover, its average *IDR* climbed by 0.02 (approximately one standard deviation), whereas patent novelty declined by 5.3%.

This finding is consistently robust across numerous methodological approaches, as shown by the comprehensive specification maps reported in Online Appendix B (Figures A.2 through A.7).[31] These epistemic maps display regressions that iterate over varying lag structures, dependent variables, and control variable combinations. Although prior research provides some guidance on variables that explain firm innovation, it provides limited direction regarding optimal lag periods, measure construction, and appropriate combinations of control variables. The epistemic maps in the

Figure 5. (Color online) Short-Termism, Firm Patent Portfolio Influence, and Novelty, by Period



Notes. The figure presents binned scatter plots illustrating how *IDR* relates to the average influence (a) and novelty (b) of a firm's patent portfolio in the underlying data. *Patent Influence* is based on the tree of direct and indirect forward citations, whereas *Patent Novelty* is based on a text-based metric relative to prior art. Individual observations in the data appear as scatter points in the background of the figure. To capture temporal shifts, these observations are split into two periods (1997–2002 vs. 2003–2009 for influence or 2003–2015 for novelty). For visual clarity, we have excluded observations with extreme *IDR* values falling outside the plotted bins. We also dropped fewer than 20 cases where the value of novelty exceeds 10 to preserve the scale of the plot. Binned scatter plots are generated without adjusting for any controls. Binned means are fitted with a line to show the relationship in the raw data between *IDR* and influence and novelty, respectively, in each period. Across periods, *IDR* (short-termism) is negatively related to both a firm's average patent portfolio influence and novelty. This negative relationship is more pronounced (negative) in the post-2002 period.

Online Appendix allow us to assess whether our key relationships remain consistent across methodological choices. Based on this comprehensive analysis, we conclude that the negative relationship between a firm's short-term orientation and technologically significant innovation remains stable regardless of these methodological decisions.

Importantly, these effects are wide-ranging and not solely driven by the right tail of the distribution. The results remain robust when using median-based influence and novelty measures, which are less sensitive to extreme values within a firm's patent portfolio compared to mean-based measures. Between-firm comparisons reinforce this pattern: binned scatter plots of the raw data consistently show negative relationships between *IDR* and firm influence or novelty after the top 5% of firms by influence or novelty are excluded.[32] Furthermore, *IDR* is associated with fewer highly influential or novel patents a firm produces (see Table A.12 in the Online Appendix), suggesting that its effect extends to breakthrough generation.

Beyond our primary analysis, we take alternative approaches to capturing the generation of technologically significant innovation in the Online Appendix. These analyses examine whether our main finding of a negative relationship between short-termism and technologically significant innovation holds even when using fundamentally different approaches to measure breakthrough innovation. First, we investigate firms' entries into nascent technological areas—potential precursors to breakthroughs characterized by high uncertainty and significant potential. Table A.15 in the Online Appendix presents evidence that short-termism is associated with fewer patenting activities in new subclasses. Second, using a network-based measure (Funk and Owen-Smith 2016), we assess whether firms facing more

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short-term pressures are less likely to develop inventions that destabilize technological trajectories—another indicator of breakthrough potential. Table A.16 in the Online Appendix (columns (1) and (2)) shows that the negative association between *IDR* and technologically significant inventions is precisely measured for the most innovative patents (95th percentile) but not for average patents, suggesting that rising short-term market pressure particularly dampens potentially path-breaking inventions rather than more incremental innovations. Finally, using the Kelly et al. (2021) text-based measure of technological importance—which identifies patents with low textual similarity to past inventions but high similarity to future inventions—we find further evidence that increasing short-termism is associated with lower technological importance at the firm level (Table A.16, columns (3) and (4)).

We now examine whether the negative relationship between a firm's short-term orientation and technologically significant innovation is causal.

4.3. IV Analysis: Is the Relationship Between *IDR* and Firm Patent Portfolios Causal?

If investors choose firms with strategies (and time horizons for returns) that match their preferences, then *IDR* is an outcome, rather than a cause, of firm strategy. That is, *IDR* may reflect a firm's own temporal orientation and strategies rather than external pressures placed on the firm by investors and capital markets. To isolate exogenous influences and identify whether the relationship between *IDR* and innovation is causal at least in part, we evaluate changes in institutional owners' post-merger investment styles that are arguably exogenous to the firm.

There are several compelling reasons to focus on the role of institutional ownership in explaining the relationship between capital market (i.e., investor) pressure and firm innovation. First, institutional ownership is increasingly influential in capital markets. Shareholdings of large financial institutions have grown from 8% to 68% between 1950 and 2006, whereas retail holdings have dropped from 92% to 32% in the same period (Bogle 2006). By 2009, institutional owners held 73% of the equity of the top 1,000 U.S. companies (Gilson and Gordon 2013, Barton and Wiseman 2014). Second, institutional owners vary in their temporal orientation, with some owners displaying significantly shorter holding periods, smaller holdings, and more diversified portfolios than others (Bushee 2001). The short-term-oriented approach of some owners contributes to rising market pressure for short-term firm behavior and ultimately affects firm decisions on projects with uncertain returns, such as longer-term R&D (e.g., Bushee 1998, Aghion et al. 2013, Troyer and Ahuja 2015). Finally, an institutional owner can shift its time orientation, and when it does, the shift operates across its portfolio and is not firm specific. Consequently, when such portfolio-wide shifts occur, the resulting impact on any individual firm can be considered exogenous to that firm's specific characteristics.

We identify significant shifts in the overall investment approach of an institutional owner by tracking whether it is viewed as a patient, long-term-oriented investor in a given year. According to data and research by Bushee (1998, 2001), institutional ownership classifications may range from "transient," a category of owners with high portfolio turnover (i.e., short-term holdings), to "dedicated," which indicates low turnover and longer-term holdings. Note that these classifications are made on the investor level, not the investor-firm level. That is, if an investor is categorized as "dedicated," this classification operates across all of that investor's holdings and is not unique to any firm in its portfolio. Given the robust findings of a positive (negative) association between transient (dedicated) ownership and the *IDR* measure (Sampson and Shi 2023), we use changes in the classification of a firm's institutional owners as an instrument for *IDR*.

To identify plausibly exogenous changes to an institutional owner's investment style, we focus on changes in classifications following mergers between institutional investors.[33] Mergers between financial institutions are plausibly exogenous to individual portfolio firms, because they are precipitated by an array of factors external to any single portfolio firm, including deregulation, merger market conditions, and the institutional investor's operational opportunities (Houston et al. 2001, He and Huang 2017). Whether and how ownership style changes postmerger hinges on the specific portfolio characteristics of the merging institutions, which are difficult to predict ex ante.

Following He and Huang (2017) and DesJardine et al. (2023), we compile the data on all the mergers of financial institutions during 1983–2011. We then identify 21 institutions that underwent mergers associated with changes in ownership styles. As recognized in these prior studies, although mergers between large financial institutions are infrequent, their implications are likely far-reaching because of the scale and influence of their portfolio decisions. The mergers involving these 21 institutions affected the institutional holdings of over 10,000 firms between 1986 and 2010 in the original Securities and Exchange Commission (SEC) 13F data. One example is Wells Fargo's 2003 acquisition of Benson Associates, which specialized in value investing, a longer-term investment approach. (Wells Fargo's pre-acquisition investment style was more transient, characterized by higher turnover.) After Benson Associates' portfolio and team were integrated into Wells Fargo's institutional investment unit, the combined entity shifted away from the transient ownership style and managed a diversified portfolio as a quasi-indexer,

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indicating a marked slowdown in portfolio turnover and rise in holding periods.

From these data, we construct two instruments for *IDR*. The first instrument, *Transition from Dedicated after Merger*, counts the number of "dedicated" institutional owners of the focal (portfolio) firm that changed their investment style following an institutional merger. Specifically, we count instances where institutional investors shifted away from the dedicated ownership style either during the same year as a merger or in the year immediately following it. This instrument captures these ownership style transitions that occurred within the two-year period preceding *IDR*. Similarly, *Transition from Transient after Merger* measures the number of institutional owners in the focal firm that shifted away from the transient ownership style following an institutional merger. This count includes institutional investors that changed their style either during the merger year or in the subsequent year. As with the first instrument, we measure these transitions over the two-year period preceding our *IDR* measurement. In constructing both instruments, we specifically focus on acquiring institutions within these mergers because of data availability constraints. Our analysis below excludes firms whose institutional owners have not experienced any mergers during the sampling period.

We expect that exogenous changes in an institution's temporal focus following a merger, as captured by the instruments, will alter the investor pressure on firms in its portfolio, as reflected in *IDR*. Specifically, the transition from the dedicated (transient) ownership style may lead to a higher (lower) discount rate of future returns, or greater (lower) short-term pressures on the firm. The validity of these instruments rests on the assumption that institutional ownership classification changes occur independently of any specific portfolio firm. This independence appears justified because classifications are determined based on an owner's behavior across its entire portfolio of shareholdings. It is difficult to imagine a situation where an institutional owner would alter its overall investment pattern because of a single firm in its portfolio, particularly because we exclude all cases in which classification changes coincide with the addition or removal of the focal firm by the institutional owner. Nevertheless, there are important potential counterarguments to this assumption, which we now consider.

Despite our above priors, managers of a focal firm might not adjust their temporal focus and strategy in response to institutional investors' investment style changes when those institutions maintain their shareholdings. This potential disconnect could occur for two reasons. First, managers might interpret the continued investment as a vote of confidence in the firm's current strategic direction. Second, they might attribute the institution's decision to retain shares to other considerations unrelated to the firm's performance or strategy. If firms indeed fail to realign their R&D strategies in response to these investor classification changes (e.g., transitions away from dedicated), this would likely reduce the effectiveness of our instruments. However, we do not find empirical evidence of a weak instrument issue. It is also worth noting that this alternative explanation assumes managers can accurately interpret institutional investors' holding decisions, which are generally difficult to decipher from the outside.

In addition, our instruments capture changes that are partially a function of broader market dynamics including the behavior of other institutions. This introduces an element of randomness that reduces concerns about deliberate selection or matching between specific institutions and firms. Nevertheless, one might question whether the exclusion restriction could be violated if there is inherent alignment between a holding institution's preferences and a firm's strategic orientation. To address this concern, we use "partial identification" techniques developed by Conley et al. (2012) (see also Frake et al. 2025) to examine whether our result holds even if exclusion restrictions are violated. This investigation confirms that the main effects of *IDR* remain robust even when the exclusion restriction is violated.[34]

Using these instruments to evaluate the causal impact of *IDR* on innovative outcomes, we estimate each measure of innovative outcomes as a function of instrumented *IDR* via a two-stage least squares (2SLS), fixed-effects regression. We provide illustrative regressions in Table 5 for all dependent variables using our instrumental variable approach described above. Through a series of diagnostic tests for the IV estimation, we find empirical support for the validity of our chosen instruments.[35]

Results in Table 5 offer support for our causal claims. For *Patent Influence*, *Patent Novelty*, and *Influence* × *Novelty*, the coefficient on the instrumented *IDR* variable is negative and precisely estimated (with p-values of <0.001, <0.01, and <0.05, respectively).[36] These effects roughly map to a single-standard-deviation increase in *IDR* leading to a 10%–13% drop in *Patent Influence* and *Patent Novelty* on average. In the first stage for each model, we find that *Transition from Transient After Merger* negatively correlates with *IDR* as expected; firms with investors that move away from short-term holding periods show declines in *IDR*. Counter to expectations, however, *Transition from Dedicated After Merger* is also negatively correlated with *IDR*.[37] The economic magnitude of these effects is substantial, with one postmerger institution transition causing at least an 18.2% standard-deviation decrease in *IDR*.

Although a full exploration of these relationships is beyond the scope of this paper, we note that the overall drop in *IDR* is consistent with a slowdown in turnover

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of institutional holdings, which reduces short-term pressures on firms, possibly because of the need for stabilization and integration following a merger (Holthausen et al. 1990, He and Huang 2017). In addition, the combination of the equity holdings of merging institutions may broaden the postmerger portfolio and transform dedicated or transient investors into quasi-indexer institutions, whose link to short-term market pressure is more ambiguous. Although quasi-indexers' holdings tend to be more diversified than those of dedicated and transient institutions, they are also characterized by infrequent portfolio turnover (Bushee 2001). Porter (1992) has argued that quasi-indexers may increase short-term pressures on firms because of their limited voice and influence (resulting from their diffused holdings). This argument assumes that firms naturally gravitate toward short-term behavior that institutional investors would otherwise temper. However, recent research counters this perspective. Sakaki et al. (2021) argue that quasi-indexers, as stable institutional owners, actually represent long-term investors and find that their investment leads to more efficient, less volatile share prices for portfolio firms.[38] Further, Liu et al. (2019) find that the more passive investment style characterized by quasi-indexer classification predicts increases in patent quantity and quality among portfolio firms.

Consequently, if postmerger shifts toward quasi-indexer classification reduce investor scrutiny and market volatility, this transition may alleviate pressure on firms, potentially benefiting their higher-risk, higher-uncertainty research projects. We leave further exploration of this intriguing finding for future research and focus here on how components of *IDR* that are exogenous to the firm affect technologically significant innovation. Overall, these instrumental variable analyses provide additional support for the robust negative relationship between *IDR* and innovative outcomes, suggesting that this relationship may be causal. Similar to our results reported in Section 4.2 above, our instrumental variable findings are robust to a wide variety of specifications and alternative dependent variables set out via the specification maps reported in Online Appendix B, Figure A.7.

4.4. Does Rising Short-Term Orientation Shift the Locus of Technologically Significant Innovation?

Our analyses above focus on the patenting activities of publicly traded firms. However, an important question remains: Could the negative relationship between short-termism and technologically significant innovation be offset by an alternative innovation strategy? Specifically, short-term-oriented public firms might compensate for reduced internal innovation by acquiring innovative start-ups. This would represent a shifting division of labor between established public firms and private, entrepreneurial companies (Arora et al. 2020), where start-ups serve as external R&D pipelines for riskier, longer-term projects that can generate technologically significant innovation. To investigate this possibility, we matched the patenting records of venture-backed companies (from PitchBook data) to mergers and acquisitions (M&A) deals where public firms targeted such VC-backed start-ups. Although these data do not capture all innovation generated by start-ups, they allow us to compare both the quantity and quality of inventions developed by start-ups that are acquired by more (less) short-term-oriented public firms. Note that tables generated for this section are reported in Online Appendix A.

Our analysis of patents by start-ups acquired by public firms, reported in Table A.17 in the Online Appendix, suggests that the division-of-labor argument has some merit: more short-term-oriented public firms acquire more patents through acquisitions of venture-backed start-ups (Model 1). These acquired start-ups tend to be relatively mature. At the time of acquisition, the median age of these start-ups is nine years, with over 89% having operated for five years or more. The median age of the target start-up patent portfolio—defined as the number of years between the patent grant and the acquisition year, averaged across all patents filed by a start-up before acquisition—is 1.33 years, suggesting that public firms are acquiring the start-ups at least in part to access their novel, recently developed technologies. These statistics suggest that start-up acquisitions offer public firms access to external R&D that delivers quicker returns and reduces uncertainty, aligning with the capital market's growing short-term focus.

Meanwhile, Models (2) and (3) in Table A.17 in the Online Appendix show that this positive association between short-term orientation and patents in acquired start-ups is not driven by the acquisition of high-quality patents as defined by top percentiles of influence and novelty. These estimates are not precise, however; we have limited statistical power because of the focus on the variance in the extreme right tails (i.e., of influential or novel patents). Still, conditional on acquisition, public firms facing more short-term pressure tend to acquire start-ups whose patents are less influential and novel on average (Models (4) and (5)).

Our analysis of patent ownership transfers among public firms, facilitated by the DISCERN project (Arora et al. 2024b), further supports these findings. We find that short-term-oriented firms tend to acquire less influential and less novel patents from other public firms, whereas the relationship between short-termism and the quantity of patents acquired lacks statistical precision (Table A.20 in the Online Appendix). These results collectively indicate that the effects of short-term orientation extend to both internal and external

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innovation strategies of public firms. Although short-term orientation has prompted public firms to increasingly acquire externally developed innovations, this strategic shift does not compensate for the declining quality of technologically significant innovation produced internally by firms experiencing rising short-term pressures.

We also conducted a descriptive analysis of whether exposure to rising *IDR* in public markets affected innovation by VC-backed start-ups, even if those start-ups were not acquired by public firms. Specifically, we identify inventions patented by VC-backed start-ups in industries where public firms experienced the largest increase in *IDR* in the 2000s and compare them to start-up patents in industries with less substantial increases in *IDR* in the same period. This represents another approach to examine whether the locus of innovation changes in response to declining technologically significant innovation by public firms. This analysis, reported in Table A.18 in the Online Appendix, indicates that start-ups patent less influential or novel inventions when their industry is exposed to a significant increase in short-term pressure from public investors, even though most of those start-ups are still private. In other words, start-up innovation efforts may be influenced by the public market, but not in a way that leads to a compensating change in the locus of technologically significant innovation. However, we cannot rule out that industry-wide declines in technologically significant innovation may be caused by declines in technological opportunity in an industry.

These results collectively suggest that the locus of technologically significant innovation is not shifting markedly between public and private firms in response to rising short-termism. Instead, our findings point to a broader pattern: short-term financial pressures appear to concentrate the development of groundbreaking innovations in fewer firms in the innovation ecosystem, regardless of whether these innovations originate within public firms themselves or are sourced through acquisitions of entrepreneurial ventures.

5. Discussion and Conclusion

We have presented three central findings in this paper. First, the share of highly influential and novel patents from public and private firms has stayed relatively stable over time. Although universities and VC-backed ventures have grown their share considerably since 1985, they still represent only a small fraction relative to public and other private companies. Second, these top-level proportions mask considerable within-group shifts: since around 2000, public firm patenting of influential and novel technologies has become increasingly concentrated within a shrinking number of firms. Lastly, among public firms, short-term orientation is negatively associated with the production of these significant, breakthrough inventions. This relationship is robust across multiple measures of technological significance, as well as a broad set of modeling assumptions.

This robust negative link between short-term focus and firms' generation and acquisition of technologically significant inventions may arise either because investors choose firms whose strategies match their own temporal preferences or because investors with short-term preferences exert external pressures on firms' strategies. To isolate investors' exogenous influences on firms and establish the causal effect of *IDR* on innovation, we employ an instrumental variable approach that leverages financial institutions' postmerger behavior changes, which are plausibly independent of individual firm characteristics and technological opportunities. The evidence from this analysis lends merit to the possibility that at least part of the relationship between rising short-term orientation and declining influential inventions within the firm is driven by market trends exogenous to firm opportunity (i.e., changes in investor preference that are unrelated to firm characteristics or opportunities). This stands apart from a technological opportunity perspective that could also explain an increasing focus on short-term goals. In this view, fundamental technological opportunities have been diminishing (Gordon 2017). Indeed, Jones (2009) documents that innovation is becoming more challenging, at least insofar as it requires bigger teams to achieve comparable breakthroughs. If search for or development of novel approaches has become more expensive, and the returns to this added expense are insufficient, then part of the shifts we document may be considered appropriate, competitive responses. By the same token, longer time horizons may naturally reemerge as new technological opportunities arise. Evidence for this exists in certain sectors where nonprofit institutions and private firms actively pursue fundamental research, such as artificial intelligence, mRNA, and their related applications. Therefore, although our findings indicate that fewer firms may be conducting fundamental innovation, those that still engage in long-term, riskier innovation projects may be doing so in highly productive ways. Our analyses reveal that, although this explanation may have merit, investor preferences regarding return timing play a causal role in shaping U.S. public firm innovation.

At the firm level, rising short-term orientation creates pressure for immediate returns from R&D investments, incentivizing managers to prioritize incremental innovations with rapid payoffs rather than more fundamental research. However, when firms underinvest in influential or novel inventions, which often generate revenue streams for decades as exemplified by the Xerox laser printer discussed above, their long-term competitive

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positions may erode because of reduced development of progenerative inventions. Although not every firm is poised to pursue novel and influential inventions and fully capture their private benefits, such inventions yield significant social welfare benefits by propelling technological progress and improving productivity across industries (Corredoira and Banerjee 2015, Arts et al. 2021, Kelly et al. 2021). The full implications of fewer public firms' focus on technologically significant innovation thus remain an important area for future research, both at the individual firm and broader economic levels.

The robustness of our results across measures of novelty and influence, as well as patent acquisitions, suggests how short-term orientation may affect various facets of firm innovation. Firms that produce fewer novel inventions likely conduct more constrained searches around established foundational principles rather than exploring new technological territories. Similarly, the reduced likelihood of producing novel innovations suggests that firms with strong short-term orientation also retreat from the extended development required to advance novel innovations. Instead, they confine themselves to incremental improvements of existing technologies rather than pursuing embryonic, high-potential innovations. Further, because short-term-oriented firms are less likely to acquire influential and novel patents, it does not appear that these firms are outsourcing their long-term-oriented R&D. An absorptive capacity logic would suggest that there are also fewer acquirers with sufficiently long-term horizons to bring early technologies to market. This carries significance, because innovations that resolve core bottlenecks and open valuable technological pathways often generate long-term, sustainable value. Our results suggest that shifting market preferences toward short-term returns can potentially create constraints on firms' strategic innovation options. Further research will be needed to understand how investor pressure for short-term returns responds to the emergence of new technological opportunities.

Beyond the impact on individual firms, our findings provide insight into the changing nature of R&D activities among U.S. public firms. An increased concentration in the production of technologically significant innovation within fewer firms, even if there are overall increases in research and development activity, may create a narrower range of ideas and approaches. Further research could investigate whether this concentration is detrimental to innovative progress, perhaps because of a dearth of unique perspectives and experimentation, as a more limited number of firms set research agendas, shape standards, or define industry norms. This rising concentration may also appear as concerns about market power. Fewer highly innovative firms often means those firms can outcompete or acquire smaller rivals, potentially reducing overall competition and future innovation from new market entrants (Cunningham et al. 2021). Such consolidation could heighten sector or geographic vulnerability as well: if one or two large companies face economic or political setbacks, this could impact entire industries, or geographic areas. These interrelated issues deserve continued study to ensure that a variety of ideas and actors can continue to fuel broad-based innovation well into the future. These patterns underscore important concerns about how market dynamics, including short-term pressures, may negatively affect the competitive diversity and resilience of the innovation ecosystem.

This observed increase in concentration merits attention, particularly given the absence of comprehensive evaluation regarding short-termism's aggregate impact on R&D outcomes (Roe 2022). Whether evolving corporate R&D boundaries and focus areas ultimately mitigate or intensify these short-termism challenges remains unclear, a question that emerging research using advanced data on patent ownership transfers and corporate scientific activity may help further address (Arora et al. 2024b).[39] Assuming efficient markets, Kaplan (2018) proposes that venture-capital-backed start-ups may fill innovation gaps left by public firms, citing transformative technologies such as fracking, internet infrastructure, and biotechnology as illustrations—all of which required substantial patience before generating returns. The phenomenon of inventor entrepreneurship (e.g., Gambardella et al. 2015), where researchers constrained by corporate short-termism may leave and establish independent ventures, may contribute to this shift. Additionally, public institutions have historically played substantial roles in shaping technological trajectories (Corredoira et al. 2018, Fleming et al. 2019), suggesting that government and university laboratories may have expanded their fundamental research activities in response to shifting corporate priorities, potentially addressing critical aspects of the R&D gap, even though the recent federal funding cuts have called this possibility into question for the future. Further, our results are consistent with, at best, only a limited shift of fundamental research capabilities from public corporations to start-ups, private firms, and universities.

Our evidence—that the growth of influential and novel patents generated by start-ups and other private firms has been limited—aligns with discussions about current entrepreneurs potentially focusing on less impactful problems (Lapowsky et al. 2018). This result is consistent with Barrot (2017), who finds that venture capitalists' investment choices are influenced by their fund's horizon and past success, with early success leading them to invest in less innovative companies. Public market conditions further constrain this

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potential compensation mechanism. It is also consistent with Nanda and Rhodes-Kropf (2013), who argue that venture capitalists will take fewer risks when they anticipate difficulties in exiting investments through public markets. A critical research question concerns whether increased concentration of firms capable of producing highly influential patents reduces acquisition (i.e., exit) opportunities for start-ups, thereby depressing start-up valuations and constraining their strategic options. The negative relationship between short-term orientation and the acquisition of technologically significant innovation from start-ups also suggests that U.S. public firms are not fully substituting for a short-term internal R&D focus by leveraging external R&D from start-ups. Thus, our findings suggest that the increasing short-term orientation of public firms may exert influence well beyond their organizational boundaries, potentially constraining innovation throughout the private equity ecosystem, particularly during periods of limited market liquidity (e.g., the cooling of the special purpose acquisition company (SPAC) market).

An important limitation of this work is the difficulties in capturing a more complete spectrum of firm innovation or fully disentangling the intricate and dynamic mechanisms in the broader R&D ecosystem with current data. It is possible that our findings on technologically significant patents do not accurately represent settings where influential inventions are more architectural or emergent in nature and thus not well captured by patent data, such as software and digital platforms. Similarly, we cannot rule out the possibility that some nonpatenting firms may have also developed influential or novel technologies. As such, a critical next step is to explore alternative indicators that can better reflect the non-patent aspects of influential innovation, provide a more comprehensive view of the increasingly complex landscape of firm innovation, and ultimately mitigate potential biases introduced by relying solely on patent metrics.

Our study is also constrained by the levels at which key empirical measurements and analyses are operationalized. Because variation in short-termism is observed at the firm level, we lack insights as to how individual projects or their components are discounted within a firm or evaluated by investors (to the extent that investors have the ability to observe and evaluate individual research projects within portfolio firms). This limitation leads to evaluation of patent portfolios in the empirical design, rather than individual innovations or components. The introduction of new data enabling researchers to identify within-firm R&D projects as an intermediate layer of the analysis would provide a more nuanced understanding of how temporal preferences and R&D outcomes interact at a more granular level. With such data, we may be better able to evaluate more nuanced firm R&D portfolio-level decisions, including whether and how firms balance the needs for short-term payoffs with longer-term bets. We leave these important questions to further study.

Our findings highlight a shift in the nature of innovation among U.S. public firms, revealing a robust negative relationship between short-term market pressures and the generation of technologically significant inventions. Firms subjected to stronger short-term financial pressures systematically produced patent portfolios that are less influential. This systematic shift toward incremental innovation represents a significant transformation in how most public firms approached innovation, with potentially important implications for their long-term competitive advantage and sustained economic growth. Furthermore, our analysis indicates that this trend has not been compensated by alternative strategies, such as acquiring innovative start-ups. Firms pressured to focus on short-term returns tended to acquire fewer groundbreaking patents, pointing toward a general weakening of the innovative output of their patent portfolios. These findings suggest that there has been a broader reorientation of many corporate R&D strategies, in part driven by investor preferences for immediate returns, raising important questions about how managers and policymakers should consider the role of capital markets in shaping the technological competitiveness of firms.

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Endnotes

[1] https://web.archive.org/web/20160308115624/https://www.bell-labs.com/our-people/recognition/, accessed December 10, 2023.

[2] https://www.nytimes.com/2015/04/10/business/dealbook/peltzs-attacks-on-dupont-threaten-americas-research-edge.html.

[3] More specifically, the National Science Foundation's (NSF's) National Patterns of R&D Resources data series suggests that the relative difference between development ("D") and research ("R") expenditures has grown considerably, showing that the difference between development and research expenditure has grown by 35.3% from the 1980s to 2015, our sample period. See also Boroush and Guci (2022) on the shift from research to development in the United States.

[4] Knott (2016) measures a decline in RQ, a measure of R&D productivity among public firms, and documents changing strategies around how innovation is generated. Corredoira et al. (2018) find in a sample of patents granted between 2001 and 2004 that those funded by the government were more likely to spark follow-on

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innovation than those solely funded by firms. Bernstein (2015) documents that firms going public changed R&D strategies to focus on greater external acquisition of new technologies. Budish et al. (2015) similarly show, in the context of cancer treatment development, that private research has moved away from long-term projects. These results, coupled with evidence that U.S. firms develop less science (Arora et al. 2018), is suggestive evidence that a shift in the nature of firm R&D has been taking place among firms.

[5] A patent has "rich forward citation trees" when it has more diverse and extensive classes and assignees in its forward citation trees (Corredoira and Banerjee 2015).

[6] Cremers et al. (2020) similarly find that firms reduce R&D spending with exogenous increases in transient investors.

[7] See, for example, Ke and Petroni (2004), Ke et al. (2003), and Bushee (2001).

[8] Gonzalez-Uribe and Groen-Xu (2017) also demonstrate that firms with CEOs having longer, fixed-term employment contracts produce more heavily cited (and presumably valuable) patents. However, it is worth noting that existing evidence is mixed on whether CEOs facing short-term pressure directly cut R&D spending (e.g., Cheng 2004). Nonetheless, corporations have exhibited a notable shift in R&D expenditure allocation, with more funding directed toward development relative to research in recent decades (National Science Foundation 2024). See also Gentry and Shen (2013) and He and Tian (2013) for more illustrations of how short-term orientation affects firm investment.

[9] In Online Appendix F, we outline arguments and empirically examine the role of managerial incentives in explaining the relationship between short-termism (i.e., *IDR*) and technologically significant firm innovation. Results are highly consistent with those reported in this paper and are suggestive of the role that managerial compensation linked to stock prices plays as a mechanism by which managers can be influenced by investor preferences.

[10] In our empirical work below we cannot directly measure whether, for example, a new innovation relies on multiple new subtechnologies that had to be created; however, we do capture how influential a new innovation is by virtue of the technological trajectories that it impacts in future as well as firm entry into new technological fields. Relatedly, behavioral economics and finance research provides additional insights into the relationship between investment time horizons and risk appetite, which would in term affect the patience to work out solutions to technical problems associated with long-term, higher-risk investments such as ambitious R&D projects. This body of work ties the feedback period duration (i.e., the period over which returns are evaluated) to investor time horizons and, importantly, the valuation of riskier investments. Fellner and Sutter (2009) and Thaler et al. (1997) demonstrate that extended feedback periods correlate with increased long-term investment among retail investors. Gneezy et al. (2003) highlight the underlying mechanism, showing that market prices for risky assets increased when feedback and decision making occurred over longer intervals, suggesting investors became more tolerant of uncertainty when evaluating performance over extended time frames.

[11] Although the long-term orientation of Xerox PARC has been controversial, the laser printer, eventually brought to market in 1978, earned over $2B in revenues for Xerox. The annual budget for PARC at that time was approximately $6M (Hiltzik 2000)—thus, it is difficult to claim this was not a win for Xerox. However, the time from conception to product was over a decade. Xerox has since scaled back the long-term emphasis of its research center.

[12] We also omit stocks traded over the counter and companies that have experienced changes leading to a CUSIP (Committee on Uniform Securities Identification Procedures) identifier alteration within the previous five years. Such changes encompass alterations in the company name, (reverse) stock splits, and reorganizations (Financial Industry Regulatory Agency 2016).

[13] An important concern is that patent data may not always be a comprehensive representation of firm innovation because of varying degrees of patentability of innovation across industries. In additional analysis, available on request, we find that the main findings are robust to the exclusion of software and information technology (IT) companies, whose digital/system innovation may entail fewer patentable elements. Results are also similar when we focus on the top 10% industry sectors in terms of the proportion of patenting firms.

[14] We use "permno" in the KPSS data set (Kogan et al. 2017) as the firm identifier for patents. We use the Compustat-Center for Research in Security Prices (CRSP) data in the Wharton Research Data Service (WRDS) to match this identifier to Compustat data via "gvkey" and to CRSP data via "cusip."

[15] See Figure A.10 in Online Appendix E for details on construction of this measure, including a visual example of the technology network and associated adjacency matrix.

[16] Our findings are unlikely to be driven by the potential biases associated with the logarithmic transformation of adding one to zero values for two reasons. First, relative to the scale of original measurements, one is a small value to add to retain longitudinal observations of patenting firms. Second, the main results are similar when we drop cases with zero values for influence or novelty.

[17] We drop all patents assigned to main classes with fewer than 30 patent applications in that year.

[18] The findings are consistent when we use the 95th percentile of influence and novelty instead of the mean, as well as traditional direct (single-generation) citation measures. Because of the sparsity of the data for some firm-year observations for which the percentiles cannot be precisely or meaningfully defined, we opt to report the mean values for the main analysis.

[19] Interestingly, the contemporaneous correlation between *Patent Novelty* and *Patent Influence* is only 0.004 (Table 3). This suggests that most patents that use novel language do not lead to dense citation trees, and most patents with dense citation trees are not those with very novel language. However, given the timing differences between the measures, we examine and find that backward-looking patent novelty has a positive and statistically significant relationship with the influence of a firm's later patent portfolios. In this sense, firms with patent portfolios that show high values on text-based novelty are more likely to generate influential patent portfolios later. These results are reported in Table A.25 in the Online Appendix.

[20] Because our measure of short-term orientation described below is observed at the firm level, we align the patent measures by aggregating patent measures to the firm-year level. Because of data limitations and our inability to observe meaningful links among patents, firm research projects, and project-level discounting applied within the firm, we cannot extend our analysis to more granular levels. We do, however, briefly discuss the implications of this level of analysis with thoughts for future research in our discussion section below.

[21] Although we consider dividends and stock prices in the main model, Sampson and Shi (2023) demonstrate the validity of the measure is not sensitive to dividend policies and alternative definitions of cash flows (such as expected earnings and stock repurchases).

[22] As noted in Sampson and Shi (2023), current price and dividends are assumed to be noisy but unbiased proxies of expectations of future prices and dividends. Biases in measurements mean that future prices and dividends are systematically under- or overestimated by current prices and dividends and thus would be

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arbitraged away. The proxies are noisy and make the estimates conservative. The authors further provide consistent trend estimates using a model of analyst forecasts that do not require these assumptions regarding expectations of future prices and dividends, albeit in a much smaller sample.

[23] For a full list of the validation tests, please refer to the online supplemental material in Sampson and Shi (2023) at (Online Appendix C) https://onlinelibrary.wiley.com/doi/full/10.1002/smj.3158.

[24] Note that the inclusion of these control variables necessarily limits our sample time frame, because not all controls are available over the full time window for *IDR*, *Patent Influence*, and *Patent Novelty*. However, allowing the sample to expand when we include fewer control variables yields similar, though marginally noisier, results in some regressions. These results are available on request.

[25] In additional analyses, we do not find evidence that *IDR* predicts future R&D intensity or logged R&D expenditure. As such, R&D investment level does not appear to be a mediating factor. Additionally, there is no clear association with financial slack, patent portfolio size, or the number of business segments in the next period.

[26] We capture *Firm Patent Portfolio Size* based on firm patent applications in a single year, but note that our results are robust to different time frames for this measure.

[27] Interestingly, we find that the effects of *IDR* on innovative outcomes are potentially weaker for firms with decentralized R&D (Online Appendix, Tables A.21 and A.22), which may be due to decentralization hindering top managers' ability to direct R&D efforts in response to short-termism pressure. For recent work on the relationship between R&D centralization and technological outcomes, see also Cummings and Knott (2018) and Eklund (2022).

[28] In analyses reported in Table A.17 in the Online Appendix, we show that, although more short-term-oriented firms acquire more patents from start-ups, these acquired patents tend to be less influential and novel.

[29] We also estimate the relationship between *IDR* and a firm's propensity to patent. Given that *IDR* would shift firms away from exploring novel approaches to problems (*Patent Novelty*) and foundational issues (*Patent Influence*), we expect that increasing *IDR* will push firms toward incremental innovations that eventually fall below the patenting threshold, thereby reducing the likelihood of patenting altogether. Indeed, we find that firms facing greater short-term pressures are less likely to patent. These results are reported in Table A.23 and the epistemic map in Figure A.8 in the Online Appendix.

[30] We report within-R^2 values. The within-R^2 values are modest compared to overall R^2 values in fixed-effects models and may be explained by the inherent noisiness of innovative outcomes as well as their unobserved correlates. We do not interpret coefficients on control variables. Hünermund and Louw (2023) argue against interpreting coefficients on control variables because of the lack of assurance that these estimates are properly identified and unbiased.

[31] We have also examined organizational, market, and technological conditions under which the associations between *IDR* and innovative outcomes are more or less salient. We find stronger negative effects of short-termism for firms that have centralized R&D organization, face fierce product market competition, or have lower research productivity (as measured by RQ). The short-termism effects are also more salient when firms work on technologies that are in complex fields, rely on scientific work, or are government funded. For brevity, these findings are reported in Tables A.21 and A.22 in the Online Appendix and discussed in Online Appendix D. Furthermore, small *p*-values for these estimates are unlikely driven by multicollinearity, as the average variance inflation factor (VIF) for pooled OLS models with the same explanatory variables and dependent variables is lower than two.

[32] These binned scatter plots that remove the top 5% of firms by influence or novelty are not reported here, but are available on request.

[33] In further analyses reported in Table A.13 of the Online Appendix A, we use an alternative approach to instrumenting for *IDR*, focusing on changes in classification of institutional blockholders, or those holding 5% or more of a firm's outstanding shares. To reduce selection bias, we exclude new block owners with no holdings in the previous year and previous block owners who have no holdings in the current year. Results are statistically significant, and we observe the same persistent negative correlation between the instrumented *IDR* and our measures of technologically significant innovation as reported in Table 4.

[34] Using the Union of Confidence Intervals (UCI) method proposed in these studies, we estimate that even if the effect magnitudes of the instruments in the second stage (which should be zero under the identifying assumption) are as large as estimates from the first stage, the coefficient estimates of *IDR* on patent influence and novelty would remain significantly different from zero.

[35] For all models, the Kleibergen-Paap rk Lagrange multiplier (LM) statistics (chi-squared > 99.01, $p < 0.001$) indicate that underidentification is not a salient issue in the estimation. Second, the Kleibergen-Paap rk Wald statistics (F statistic > 226.19) surpass most critical values supplied by Stock and Yogo (2005) and reject the null hypothesis of weak instruments. The null hypothesis is that the instruments are irrelevant in the first-stage (i.e., *IDR*) regression. We then test whether the instruments are correlated with the error term in the second-stage equation. Results of these tests, specifically the Hansen *J*-statistics ($p > 0.5$), suggest that orthogonality conditions do not appear to be violated, as required for instrument validity (Baum et al. 2007). The Davidson-MacKinnon test ($p > 0.3$ for *Patent Novelty* and *Influence* × *Novelty* models; $p = 0.085$ for the *Patent Influence* model) indicates marginal difference between the IV estimates and the estimates of a naive model for the influence measure only.

[36] In addition, we also collected data on the deletions and additions in the S&P 500 index and explored using these index membership changes as an alternative set of instruments (Aghion et al. 2013). Although index changes offer some explanatory power over *IDR*, they do not have sufficient strength as standalone instruments. Similarly, limiting the ownership style changes to blockholder mergers results in too few cases for our analyses.

[37] Our examination of the underlying data suggests that this is likely due to an institution shifting away from the dedicated style temporarily in the merger year and transitioning back to it afterwards. We further probe the dynamic effects of these transition events on the *IDR* of the firm from five years before to five years after. Consistent with the first-stage results of the IV estimator, we find that the effects of the total number of postmerger transitions from dedicated and transient ownership styles, as identified by the instruments, only double in magnitude and become significantly different from zero in the first two years of the post-period. In contrast, the estimates show no clear evidence of a pretrend, reinforcing the plausibility of the exogeneity assumption. Furthermore, the Kolmogorov-Smirnov test and Wilcoxon rank-sum test, which do not rely on parametric assumptions, indicate significant differences in the distributions of patent influence and novelty in the sample before and after the transitions captured by the instruments ($p < 0.01$).

[38] Sakaki et al. (2021) find that quasi-indexer institutional investors, as stable owners, reduce mispricing in equity, which is one of the possible determinants of the discounting of a firm's future cash flows and assets beyond the firm-specific risk premium (discussed in Section 3.3 above). In this sense, quasi-indexer ownership may reduce short-term pressures on firms and lower *IDR*.

[39] Please see Section 4.4, where we explore whether potential changes to the innovation ecosystem and the locus of technologically significant innovation could explain our findings. As part of this investigation, we have conducted initial explorations using the DISCERN 2.0 data and report results in Online Appendix A. Further examination of a potentially changing innovation ecosystem using patent ownership changes is a fertile area for future research.

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